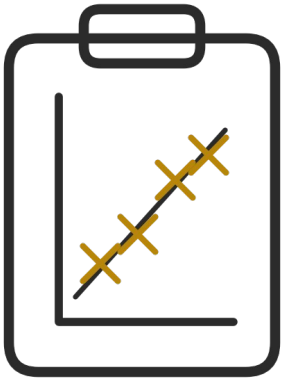
Advanced Higher Physics

Project: mark it yourself



A Scottish Borders Council resource, curated by R Stewart

Remember

This pack is for practice **before** you write your report. It is never used while you write it.

Your tick list is the official **Instructions for candidates** in the Advanced Higher Physics coursework assessment task.

Read this on your iPad. Write your answers on the printed **worksheet**.

Contents

Part 1: How to use this pack

Part 2: The marks and the tick lists

Part 3: Script 1: Determining wavelength

Part 4: Script 2: Measuring g with pendulums

Part 5: Script 3: Surface tension of water

Part 6: Answers

1 How to use this pack

Three real projects are in this pack. They come from Qualifications Scotland's Understanding Standards website. A marker has already marked them.

Your job: mark them yourself first. Then compare your marks with the marker's.

Steps

Step 1: **Read** the marks and tick lists in part 2.

Step 2: **Read** one script: part 3, 4 or 5.

Step 3: **Fill in** that script's grid on your worksheet.

Step 4: **Complete** the focus tasks on your worksheet.

Step 5: **Check** your marks against the answers in part 6.

Why marking works

The course reports advise this kind of practice. When you mark, you see what the marker looks for. You learn the standard, not someone else's words.

Do not copy sentences from these scripts. Your report must be your own work, about your own experiments.

For teachers

Use the pack during the course and in the research stage. Collect the worksheets in before the report stage.

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2 The marks and the tick lists

The project is worth 30 marks. Most marks are lost in the first five sections below.

Item	Section	Marks
4b	Analysis	4
2	Underlying physics	4
5c	Overall evaluation	3
4c	Uncertainties	3
5b	Evaluation of procedures	3
3c	Complexity and demand	3
3a	Diagrams and apparatus	2
3b	How the apparatus was used	2
5d	Quality project	1
6b	References	1
1	Abstract	1
4a	Data	1
5a	Conclusion	1
6a	Structure	1

Sections ranked by the share of marks lost in the Scottish Borders, 2024 to 2026.

Tick lists from your Instructions for candidates

Underlying physics: derive any relationships you use · define abbreviations and symbols · use terms accurately and explain ideas clearly · include diagrams, as appropriate · cite and list at least three sources.

Graphs. When drawing a graph, you must ensure that:

- it is large enough to allow the accuracy of plotting to be checked
- scales are chosen so that the plotted points are widely spread
- it has major and minor gridlines
- each axis is labelled with the name of the quantity and the correct unit
- data points are plotted accurately
- a line of best-fit is drawn, where appropriate

Evaluation of procedures (5b). You could consider:

- the accuracy and precision of experimental measurements
- the adequacy of the number of repeated measurements
- the adequacy of the range over which variables are altered
- the adequacy of control of variables

- limitations of equipment
- the reliability of methods
- sources of uncertainty

Overall evaluation (5c). Emphasise positive aspects relating to the procedures, as well as commenting on:

- selection of procedures
- problems encountered during planning
- modifications to planned procedures
- interpretation and significance of findings
- suggestions for further improvements to procedures
- suggestions for further work

Subtitles

Use the 5b and 5c items as **subtitles** in your Discussion. Only use a subtitle when you have something real to write under it. Keep your daybook under the same subtitles as you go.

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3 Script 1: Determining wavelength

This project used three procedures: electron diffraction, double slit and Newton's rings. It is 21 pages long.

Read the whole script first. Then **fill in** grid 1 on your worksheet.

Candidate 1 evidence, Physics Advanced Higher Project 2023-24, Understanding Standards. © Qualifications Scotland (SQA).

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Candidate 1 evidence

Determining Wavelength

3,529 words

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Experiment 3: Newton's Rings	page 14.
Conclusion	page 19.
References	page 20.
Appendix- Laser Graphs	page 21.

Abstract:
An investigation into the principles of wave-mechanics, specifically those dictating wavelength of EM and particle-waves.

Experimentally this is to be achieved by determining the wavelength of EM radiation (or other wave-type emissions) under conditions where certain quantum-level effects become noticeable on the macroscopic scale.

Three separate experiments to ascertain the wavelength were undertaken: an experiment with the diffraction of charged particles from an electron-beam; another demonstrating Young's Double-Slit Diffraction with the synchronised phase-beam from a laser; the third involving Newton's Rings as a further demonstration of photo-interference.

The wavelength of the electron-beam was calculated as:

3.7nm

The wavelength of the laser-beam was calculated as:

590nm

The wavelength of the light used in Newton's Rings was calculated as:

872nm

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Physics Principles:

Wave-particle duality is the fundamental principle that under certain circumstances, particles can exhibit certain properties traditionally associated with waves under the classical perspective on particle-wave theories of light.

Although this principle has been experimentally established to function with larger compound particles up to atomic and molecular scale, over the course of this investigation the principle was proved with more elemental emission types: simple light as EM radiation (both laser-synchronised & diffuse); and electrons exhibiting duality.

Particles- in this case, electrons- can exhibit wave-like properties, namely diffracting as they pass through an electromagnetic barrier. As electrons and other leptons are charged particles that interact with baryonic matter through means of Coulomb forces, the degree to which their wavelength varies relative to other particles and EM waves; this effect is calculated as the Atomic Form Factor. This factor influences the wavelength of the electrons, given by the de Broglie equation:

$$\lambda = \frac{h}{p}$$

…whereby h is Planck's Constant, (recognised as 6.63×10^{-34} Js); and p constitutes the velocity of the electron, which may be sufficiently close to the speed of light to induce relativistic effects. [1]

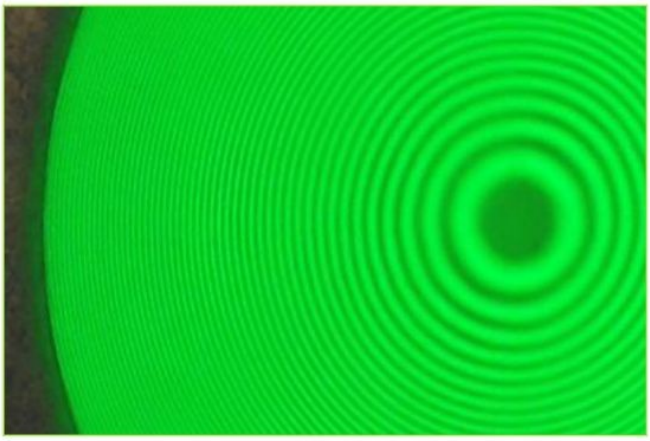
Due to this principle, it is possible to calculate the wavelength of ordinary baryonic particles in addition to massless photons, even including leptons such as electrons. Accordingly, once the wavelength of these 'matter-waves' is known, the principle can be applied to predict the way matter-beams can behave as waves- with wave-like properties such as diffraction, refraction, and (as tested here) interference.

In the case of electron-beams, the formula for the diffraction of the electrons as a matter-wave is:

...where ϑ is the angle of incidence, d is the separation between the grating through which the beam is projected, and m refers to the number of the wave-maxima.[2]

$$\lambda = \frac{d \sin\theta}{n}$$

At each maxima, the two matter-waves from the electron-gun will be in phase and undergo constructive interference, producing a point where the waves have a higher amplitude. In particle terms, this is expressed as an area where the particles in question impact more frequently. Evenly spaced between these maxima are points where the waves are precisely out of phase and undergo destructive interference, resulting in rings of darkness.

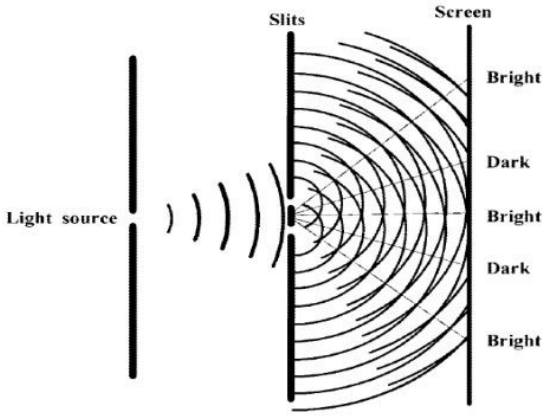


This effect is equally valid with linear spread (through a double-slit diffractor onto a flat screen) or radial spread (through an even grille onto a curved concave surface), and with massive or massless particles, as both can act as waves and produce the same set of glowing maxima. (Of course, as the impact from an individual electron is far too small to see, the effect visually appears closer to a set of smoothly glowing 'rings' of interference, as depicted left.)

If it is possible to measure the interference patterns from a matter- or energy-wave through a known slit-separation, the wavelength can be calculated and compared with a 'known' value. The formula in the case of double-slit diffraction against a flat planar surface:

$$\sin\theta = m\lambda d.$$

…where sin ϑ is the sine of the angle between each maxima, d is the slit-separation distance, and λ the wavelength of the beam.[3]

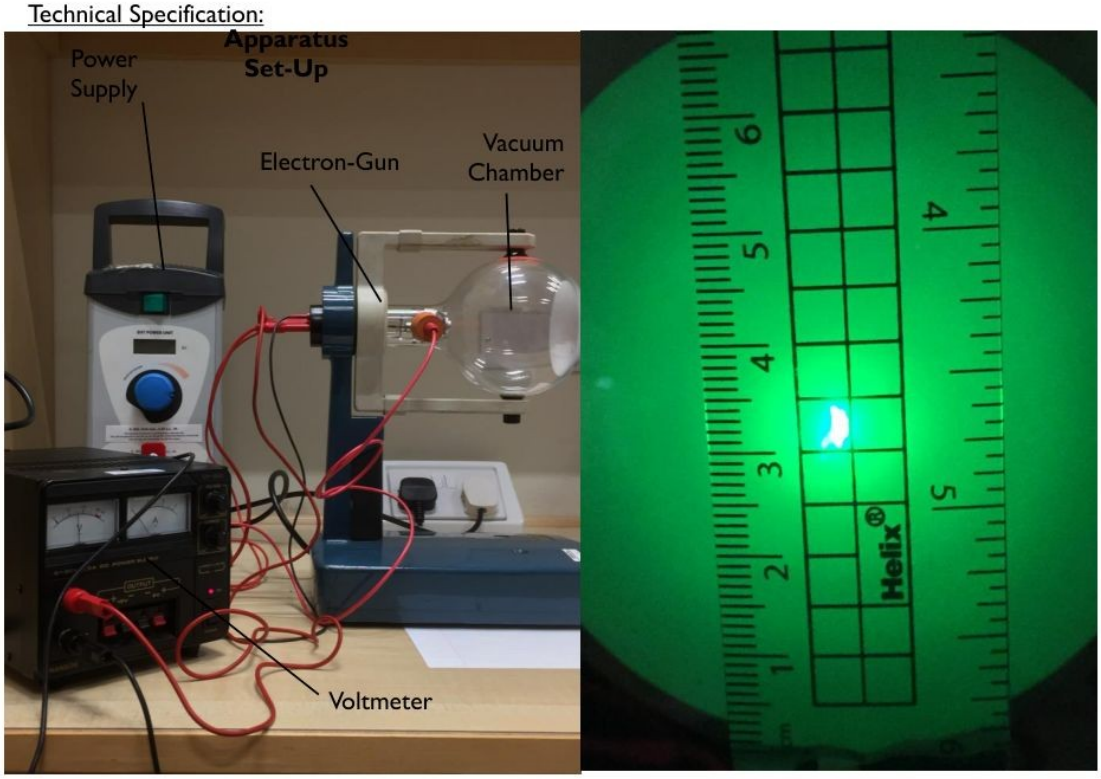


[4]

The objective, therefore, of the following experiments is to measure relevant variables relating to the behaviour of both classical 'waves' and particles exhibiting wave-like properties under duality, so as to ascertain values for the wavelength in each case. If the wavelength calculated from these experimental measurements corresponds with a 'known' value (either from the apparatus of emission, or else from de Broglie calculation), then the principle of wave-particle duality is demonstrated through the prediction and observation of wave-like behaviour.

Experiment I: Electron-Beam Diffraction

Technical Specification:



FILAMENT VOLTAGE (Vf)	---	6.3V ac/dc (8V max)
ANODE VOLTAGE (Va)		---2500-5000V (6000V max)
ANODE CURRENT (Ia)		---0.15mA @ 4000V (0.2mA max)

Experimental Procedure

An electron-diffraction tube (Telford Electronics model 555) had a carbon-screen grating inserted and was connected to a 6V AC power-supply. The heater was switched on, after which the temperature-gradient was given a minute to stabilise. The E.H.T. was set to 4kV and the lights in the room darkened.

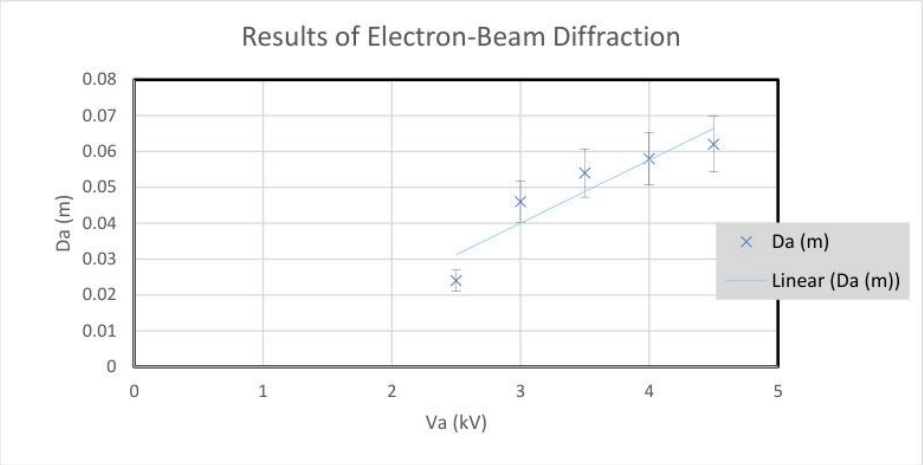
Glowing green rings around a central spot were observed, the radius of the inner ring being roughly of the mathematically-derived theoretical value. There was an inverse relationship between the anode voltage and the diameter of the ring.

Once the apparatus has stabilised in temperature, the electron-beam power-supply was set to the desired voltage- this being measured through the voltmeter in volts. The radius of the glowing green ring was then measured five times with a ruler- the distance being taken in meters- for the purposes of accuracy before moving on to the subsequent value.

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Results-

Va (kV)	Va $^{-1/2}$ (V)	D1 (m)	D2 (m)	D3 (m)	D4 (m)	D5 (m)	**Da (m)**
2.5	0.0200	0.025	0.023	0.026	0.021	0.023	**0.0240**
3.	0.0183	0.046	0.048	0.045	0.045	0.041	**0.0460**
3.5	0.0169	0.053	0.056	0.053	0.052	0.054	**0.0536**
4	0.0158	0.058	0.057	0.057	0.058	0.059	**0.0578**
4.5	0.0149	0.062	0.060	0.064	0.063	0.063	**0.0624**
5	0.0141	0.069	0.069	0.071	0.069	0.070	**0.0690**



As we know, the de Broglie wavelength of a particle- in this case an electron- is-

$\lambda = h/mv$

The velocity, *v*, can be derived as:

$eVa = \frac{1}{2} m v^2$

…which is then substituted into the de Broglie equation, giving us:

$\lambda = h/mv = h/\text{sqrt}(2emV_a) = 1.23 V_a^{-1/2}$

…finally, we can substitute in the inverse relationship for Va, producing a final equation:

$\lambda = 1.23 \times (1/0.0176\text{-}0.0128)^{-1/2}$ m

In conclusion, here we have calculated a final value for the wavelength of the electron as:

$6.63\times10^{-34}/\text{sqrt}(2 \times 9.11\times10^{-31}\times 0.0176 \times 10^{3})$

$= 3.702 \times 10^{-12}$ m.

i.e. 3.7nm

This, therefore, is the experimentally-derived value.

The calculated value of the wavelength one would expect for an electron with a kinetic energy of 2500V is 0.572×10^{-12} m, e.g. 0.572nm. Although this is an average value and therefore will not be applicable to all electrons travelling at different voltages (as seen in the experiment), it is still noticeably off our calculated value by a factor of over 5.3, or 533%.

Uncertainties:

Random uncertainties (max-min/no.of values) in the values of D are as follows:

Va (kV)	Random uncertainty in D (m)
2.5	0.001
3	0.0014
3.5	0.0008
4	0.0004
4.5	0.0008
5	0.0004

A scale reading uncertainty exists for both measurements:

V ± 0.1kV
D ± 0.0005 m

No information could be found about calibration uncertainty.

Evaluation-

There are a number of respects in which the experiment suffered from inferiorities that could have interfered with the accuracy and precision of the calculated result.

One issue experienced was the difficulty in accurately measuring the diameter of the luminescent interference rings. As electron-diffraction occurs over the diameter of the tube, the rings themselves became very blurry and lacked distinct outlines, as can be observed in the photography, making it difficult to measure an exact diameter with the ruler.

Although taking five repeated measurements per voltage doubtless helped to mitigate this issue to an extent, the degree of subjectivity involved may well have contributed to the unreliability of the data-points and final gradient.

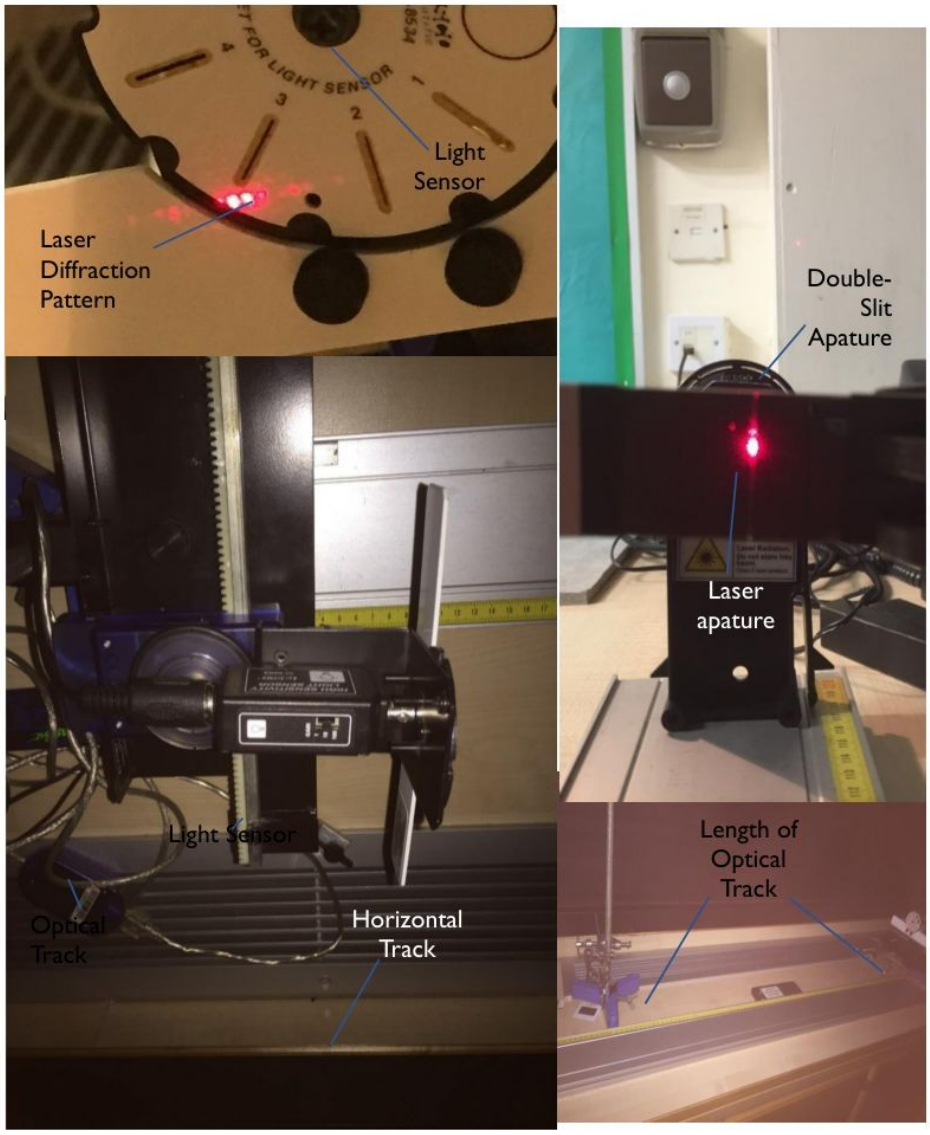
The range of variables involved was most probably a lesser issue, as the maximum voltage possible would have been 6.3 volts, and five measurements still gave a workable set of averages. Extending the voltages to which measurements were taken to 6 or 6.3 volts would have extended our data set by over 20%, but trying to take more measurements at shorter intervals of voltage may have proved counter-productive in light of the accuracy issues imposed by the vagueness of the rings, as it would have become even more difficult to observe- let alone measure accurately- subtle differences in ring diameter. Random interference (or, for that matter, the placebo effect!) would likely have proved a greater problem than the benefits brought by more readings.

Control over the independent variable was not much of an issue in this experiment. The only adjustment that needed to be made was the supply voltage, and as the mains AC was stable and the gauges perfectly easy to read, there is no reason to suppose that the supplied voltages were inaccurate. However, it is plausible to assume that a small portion of the supply voltage was lost due to inefficiency and inherent resistance in the electron-gun. Therefore, the actual voltage conveyed to the electrons may be slightly lesser than these calculations indicate.

I am doubtful that equipment limitations contributed hugely to the results of this experiment. The electron-gun, although potentially inefficient, did not appear to waver or be inconsistent in its production of the interference-patterns and there is no indication that the supply-voltage was inconsistent. I suppose that the ruler may have a small measurement uncertainty, but this is almost certainly overwhelmed by the vagueness of the rings.

Perhaps regrettably, I cannot think of a way to better de-alienate the exact diameter of the rings with the equipment we have available. I suppose, given the theoretical availability of extremely precise digital sensors, that the experiment could be conducted over a far smaller diameter vacuum-tube, thereby reducing the distance over which the electron-beams would diffract, but as this would make the luminescent rings themselves extremely small, the sensors would need to be extremely accurate to measure them precisely.

Experiment II: Young's Double-Slit Diffraction (Laser Variant)



Procedures-

The aim of this experiment was to measure the wavelength of the light from the laser-beam, by means of projecting it through a fine grille so as to induce double-slit diffraction and produce an interference pattern, from which slit-separation can be measured and the wavelength calculated.

The laser was installed on one end of the optical-track and the first of the grilles to produce the slit-separation effect fixed in front of it. Each grille had been labelled with its effective slit separation on it, and these readings were accepted as the independent variable.

At the other end of the track, an electronic light-sensor was affixed to a horizontal track travelling perpendicular to the optical track, and then plugged into a computer running the PASCO Capstone software. The distance between the grille and the front of the light-sensor was then measured with the scale on the optical track itself, which for the purposes of this demonstration has been theorised to possess the same accuracy as a ruler, and in the event this was determined to be 0.82m. This distance was kept the same over the course of the experiment; care was taken to ensure this between repeats and exchanging each grille for the next.

The laser was then switched on after Health and Safety precautions (such as installing warning signs on adjacent surfaces, mitigating the risks of potential laser-reflection, and ensuring the beam was not firing toward anything of consequence) had been taken into consideration. At this point, a luminous interference-pattern was observed being projected from the laser. The apparatus was adjusted slightly so that the pattern fell across the path of the light-sensor on the track.

The light-sensor was pulled across the path of the interference-pattern on the horizontal-track. The light-sensor was then switched-on and the PASCO Capstone software set to record and save its measurements electronically. The light-sensor was then pushed slowly along the horizontal-track so as to pass under the complete interference pattern. For each measurement, the result produced was a complicated line-graph corresponding to the intensity of the light (with each peak representing a bright maxima and each trough corresponding to a dark minima).

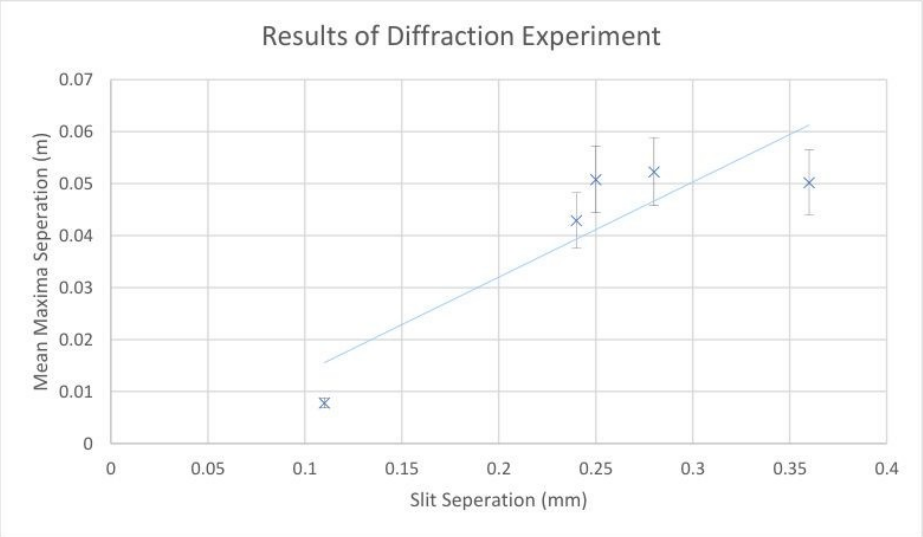
This procedure was then repeated five times for each grille. After each set had been completed, the grille was exchanged for the next one, with its slit-separation also being noted.

Each graph (see appendix for an example) was then subjected to simple analysis whereby the width of the overall pattern was divided by the number of clear peaks on the line, so as to calculate the maxima-separation. These results were then processed as follows-

Results-

Slit Separation (mm)	0.11	0.24	0.25	0.28	0.36
Maxima Separation #1 (m)	0.00750	0.0438	0.0546	0.0636	0.0630
Maxima Separation #2 (m)	0.00750	0.0460	0.0545	0.0514	0.0500
Maxima Separation #3 (m)	0.00813	0.0417	0.0571	0.0411	0.0425
Maxima Separation #4 (m)	0.00775	0.0414	0.0423	0.0529	0.0525

Maxima Separation #5 (m)	0.00813	0.0415	0.0457	0.0523	0.0430
Mean Maxima Separation (m)	**0.00780**	**0.0429**	**0.0508**	**0.0523**	**0.0502**



m = 0.042m / 0.25mm = 169.592
λ = 1/(169.592 x sin(1.571 × 10^{10} rad)) = 0.589652 x 10^{-12} m

$$\lambda = \frac{d \sin \theta}{n}$$

In conclusion, this gives us a final measurement as per

as- λ = **589.650nm**
or- **590nm** (3sf)

This is not entirely consistent with the stated value of the wavelength of the laser, which is given as 650nm (being in the red portion of the spectrum). It is, in fact, about 60nm (or 9.81%) off.

Uncertainties:

Random uncertainties (max-min/no.of values) in the values of maxima separation are as follows:

Slit separation (m)	Random uncertainty in maxima separation (m)
0.11	0.000126
0.24	0.00092
0.25	0.00296
0.28	0.0045
0.36	0.0041

No calibration uncertainty could be established and slit separation was assumed correct.

A scale reading uncertainty existed for the measurement of slit separation = ±0.00005m

Evaluation-

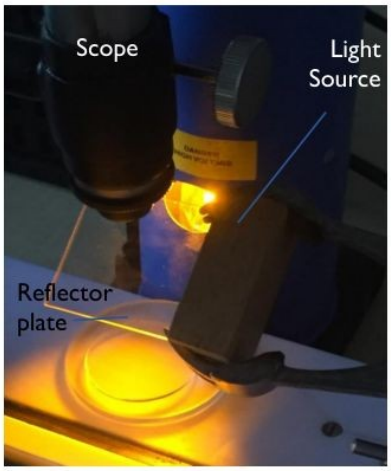
Although this experiment appears reasonably successful, there is still a slight discrepancy between the calculated value (which would be more orange or yellow in hue) and the wavelength of the light itself. Accordingly, a number of potential reasons could exist for this inaccuracy.

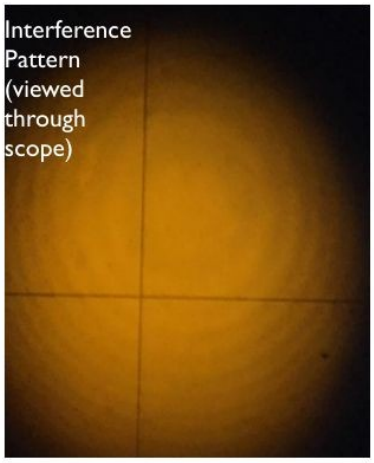
The experimental measurements, being taken as they are by a computer-controlled electronic light-sensor, are probably fairly accurate. One issue that could affect the accuracy of the results is the stability of the angle upon which the horizontal track rests on the optical track. If the horizontal track were to become slightly slanted, the light-sensor might slip outwith the interference-pattern at one end, resulting in the intensity of the measured light-pattern becoming asymmetric. If a maxima or minima were to be missed by this problem, it could interfere with the accuracy of the results, although taking multiple measurements and averaging them should help to mitigate this problem to a significant extent. As care was taken to ensure that the distance along the length of the optical-track between the grille and the sensor remained the same at all points, this should prove less of an issue compared to the risk of angular displacement.

Each measurement per slit-separation was repeated five times, which should be enough to ensure that any risk of individual anomalous measurement (or calculation, as each graph's slit-separation had to be painstakingly calculated by hand) affecting the mean unduly remains low.

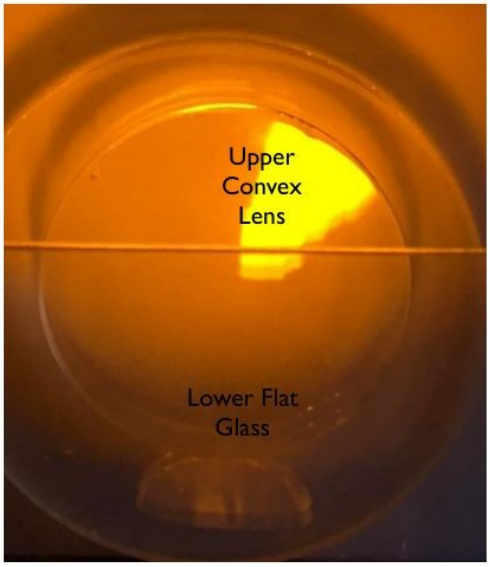
The slit-separations vary over a 0.25mm range, which is probably reasonably sufficient to ensure a good variety of result. However, as can be observed from the above graph, the final measurement (of the 0.36mm grille) is not significantly higher in terms of slit-separation than the preceding 0.28mm grille. It is possible that either the grille is labelled inaccurately, or else that the great intensity of the brightness of the light through the final grille overloaded the light-sensor (as can be observed on the final set of graphs) and produced anomalous results.

Experiment III: Newton's Rings


Scope
Light Source
Reflector plate


Interference Pattern (viewed through scope)


Horizontal Scope Track


Upper Convex Lens
Lower Flat Glass

Procedures-

The aim of this experiment was to demonstrate the principle of interference through the process of refractive diffraction. Much as the preceding two experiments, monochromatic light produces an interference pattern known as "Newton's Rings", the diameter of which can then be used to calculate the wavelength of the incident light.

The equipment was set up with a sodium lamp reflecting off a semi-silvered glass plate set at a ninety-degree angle, so as to reflect the light into the apparatus of a single convex lens balanced atop a flat glass plate. This created the interference-pattern between the two glasses and the circular air-wedge between them, resulting in the aforementioned pattern of interference.

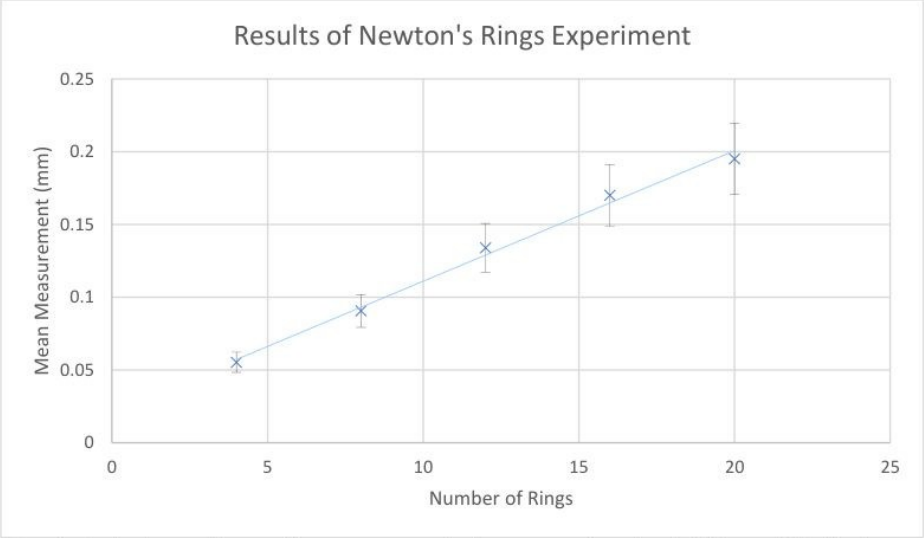
This pattern was observed through means of an optical scope mounted on a horizontal track running perpendicular to the path of the light and pointing directly down at the two glasses so as to capture the pattern. The scope was laboriously pointed at the centre of the interference pattern before measurements were taken.

The scope was then moved horizontally along the track to the extremity of each indicated dark 'ring' of destructive interference- in effect, each is a circular minima. The diameter of each ring was then measured from the centre-point by means of the vernier scale on the horizontal-scope track, allowing highly precise (if not necessarily *accurate*) measurements to be made in millimetres.

In order to counteract interference caused by any inherent asymmetry of the lens, it was decided to take measurements of the diameter of each successive ring five times at both its leftward and rightward extremities, at the fourth, eighth, twelfth, sixteenth, and twentieth rings. These results were then collocated in the following table, and means calculated for each ring.

Results-

Measurement (mm)	4	8	12	16	20
L1	0.050	0.092	0.133	0.169	0.192
R1	0.045	0.098	0.135	0.167	0.188
L2	0.061	0.085	0.131	0.172	0.196
R2	0.062	0.089	0.153	0.167	0.209
L3	0.046	0.093	0.115	0.175	0.199
R3	0.065	0.095	0.135	0.170	0.193
L4	0.054	0.087	0.135	0.168	0.196
R4	0.053	0.089	0.148	0.172	0.197
L5	0.055	0.086	0.137	0.174	0.194
R5	0.062	0.092	0.122	0.166	0.184
Mean Measurement (mm)	**0.0553**	**0.0906**	**0.134**	**0.170**	**0.195**



Conclusively, from this gradient, we can calculate a wavelength of **872nm**. This is also rather anomalous, as one would expect the yellow-coloured sodium light source to be somewhere in the wavelength region of 570nm-520nm. This results in an excess wavelength of around 40-45%.

Uncertainties:

No uncertainty in number of rings.
No calibration uncertainty could be established.
A scale reading uncertainty exists in the measurement of radius of ±0.0005m

Random uncertainties (max-min/no.of values) in the values of radius are as follows:

No of rings	Random uncertainty in radius to (m)
4	0.002
8	0.001
12	0.0033
16	0.0009
20	0.0025

Evaluation-

Unfortunately this experiment was racked with a number of issues regarding the accuracy and hence reliability of the results. The central issue at hand was that it proved extremely difficult not only to focus the optical set-up correctly so as to produce a clear set of rings centred precisely on the scope, but also to move the scope from side-to-side with the precision necessary (minute fractions of a millimetre) to focus on incredibly small sets of minima.

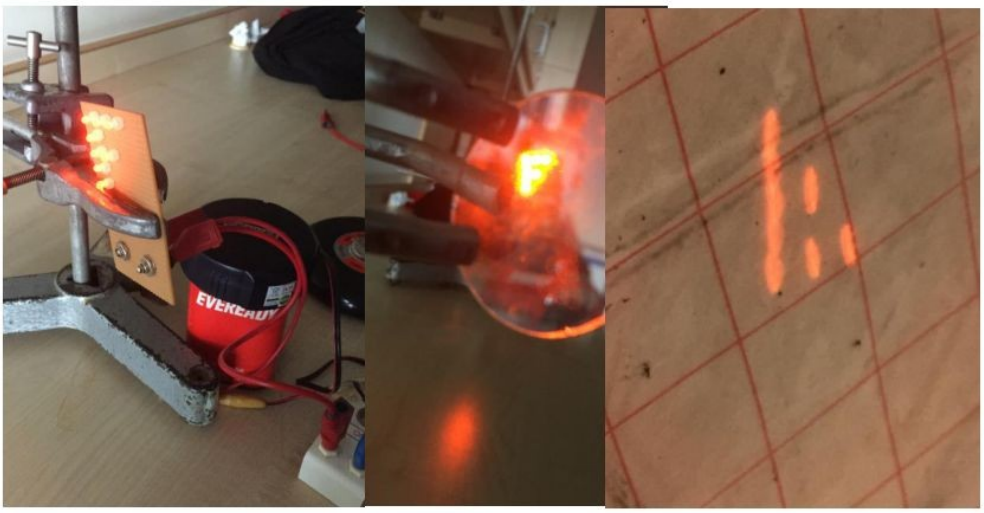
Whilst the vernier scales provided enormous precision, it was extremely difficult to maintain accuracy whilst using then and this is regrettably reflected in the large random

uncertainties displayed in the results. It may potentially prove easier to make measurements of this precision digitally using machine-heuristics or other forms of image-analysis software.

The number of unique rings measured began to prove an increasing issue toward the end of this experiment, as the rings become progressively closer-spaced the further out they travel. This is reflected in a further increase in random uncertainty, particularly toward the later rings.

One positive remark that can be observed with regards to uncertainty is the procedure of taking additional measurements to both sides of the rings, effectively doubling the number of results obtained. Not only did this have a substantial impact on reducing the damage caused to the accuracy by the occasional anomalous reading (thereby reducing random uncertainty) but also helping to counteract the issues caused by the substantial asymmetry of the lens.

The most critical issue experience with this experiment, however, was undoubtedly the great asymmetry of the convex lens used. Although not apparently obvious whilst placing the lens on the glass, and not necessarily adversely affected the results due to the two-sided method of averages used, after the experiment it was determined that the lens was, in fact, strongly asymmetric, needing to be held at a thirty-five degree angle in order to focus properly, as can be evidenced by the following photographs of subsequent attempts to rectify this situation:



Whilst the effects on the geometry of the rings themselves was compensated for by the double-sided weighing approach, the effect of the asymmetry did result in the rings being physically positioned vastly further away from the geometric centre of the lens than might have been expected, leading to the unnecessary and undesirable expenditure of a lot of time and energy chasing down the cause of a precision issue initially attributed to human error. In retrospect, this portion of the experiment would have been far better for the presence of superior optics.

Conclusion:

The objective of this set of experiments was to investigate the principles of wave-mechanics, specifically by investigating the wavelength of different particle and EM waves undergoing interference in different contexts.

The wavelength of the electron-beams undergoing diffractive interference in the vacuum-tube was calculated as **3.7nm.**

The wavelength of the laser undergoing Young's double-slit diffraction was calculated as **590nm.**

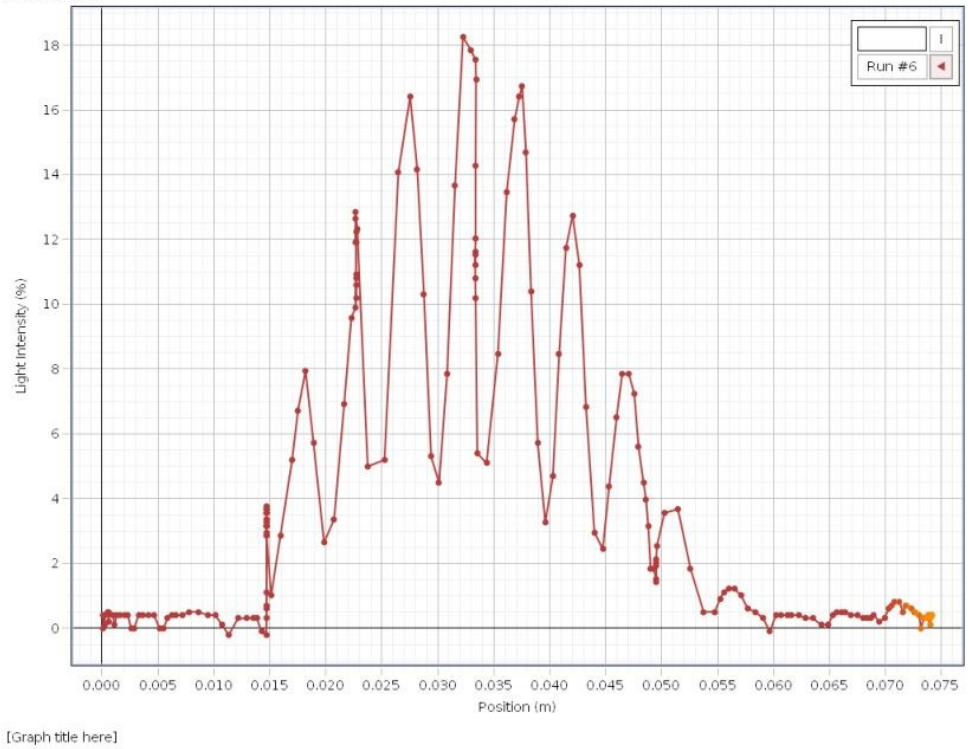
The wavelength of the sodium-lamp-light undergoing refractive interference in the Newton's Rings experiment was calculated as **872nm.**

References:

1. Horsch, W., n.d. De Broglie Wave | Physics. [online] Encyclopedia Britannica. Available at: <https://www.britannica.com/science/de-Broglie-wave> [Accessed 23 March 2020].
2. Physicsnet.co.uk. n.d. Diffraction – Single Slit, Diffraction Pattern, Diffraction Grating, Order Number. [online] Available at: <http://physicsnet.co.uk/a-level-physics-as-a2/waves/diffraction/> [Accessed 23 March 2020].
3. Ling, S. and Sammy, J., 2020. 4.4: Double-Slit Diffraction. [online] Physics LibreTexts. Available at: <https://phys.libretexts.org/Bookshelves/University_Physics/Book%3A_University_Physics_(OpenStax)/Map%3A_University_Physics_III_-_Optics_and_Modern_Physics_(OpenStax)/04%3A_Diffraction/4.04%3A_Double-Slit_Diffraction> [Accessed 23 March 2020].
4. Opentextbc.ca. 2020. Young'S Double Slit Experiment. [online] Available at: <https://opentextbc.ca/physicstestbook2/chapter/youngs-double-slit-experiment/> [Accessed 23 March 2020].

Appendix- Example of a Laser Graph

Table of Results (0.24mm)
0.04375m



[Graph title here]

End of script 1. **Fill in** grid 1 on your worksheet now.

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4 Script 2: Measuring g with pendulums

This project used three procedures: simple, bifilar and compound pendulums. It is 22 pages long.

Read the whole script first. Then **fill in** grid 2 on your worksheet.

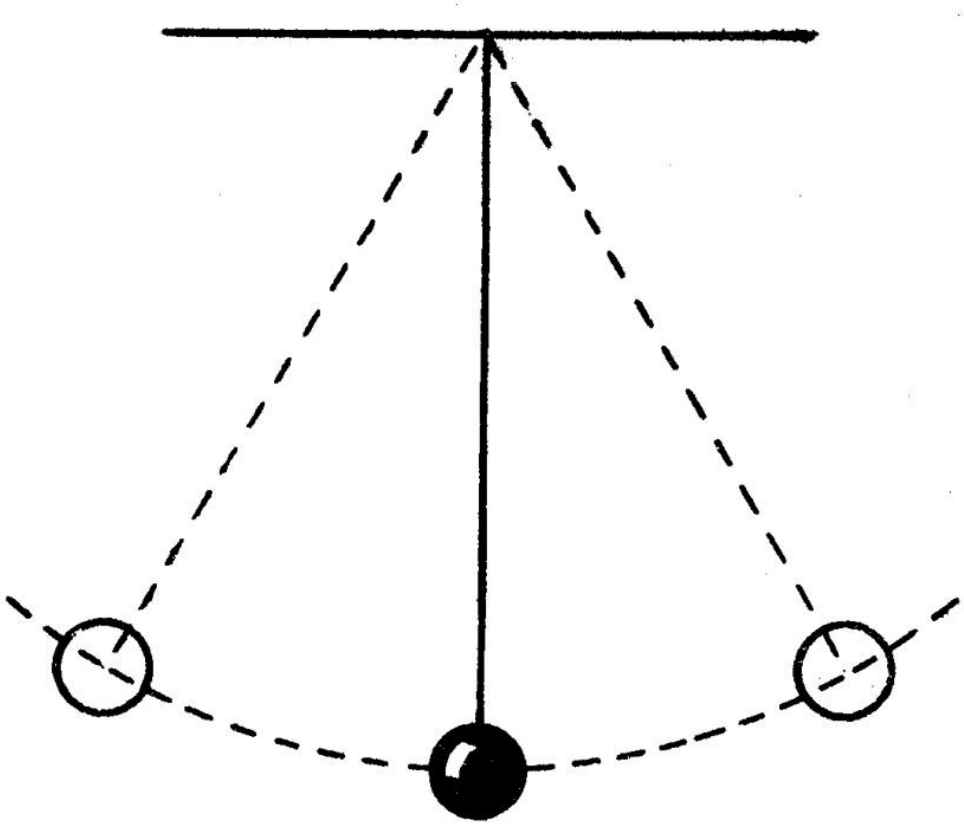
Candidate 2 evidence, Physics Advanced Higher Project 2023-24, Understanding Standards. © Qualifications Scotland (SQA).

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Candidate 2 evidence

Physics Project - Pendulums

To measure the constant g using conventional pendulums



Scottish Borders Council resource · curated by R Stewart - November 2026 · © R Stewart Physics, Electronics and Engineering

Abstract
Throughout these experiments the ultimate goal was to find a value for g using multiple different pendulums and methods. The first experiment involved using a Simple Pendulum proved to be the most accurate giving a value for g as $9.86 \pm 0.11 ms^{-2}$. For the second experiment a value of $11.1 \pm 1.7 ms^{-2}$ was found using a Compound Pendulum. This experiment suffered greatly from a lack of data points but ultimately the value found was accurate given the circumstances of the experiment. Finally, using a Bifilar Pendulum an accurate result of $10.00 \pm 0.26 ms^{-2}$

CONTENT

Underlying Physics of pendulums pg.3

Experiment 1 – Simple Pendulum pg.3

Experiment 2 – Bifilar Pendulum pg.9

Experiment 3 - Compound Pendulum pg.16

Overall Evaluation pg.21

Bibliography pg.22

Overall underlying Physics

As a rule, every pendulum that is going to be described can be said to be undergoing simple harmonic motion[1]. We understand that every object that is vibrating in this way has a restoring force being acting upon it. The restoring force that is present in all the experiments present are caused by its weight. This means that displacement of weight through each pendulum will affect the oscillation accordingly. Fundamentally, this is the difference between pendulums.

Simple pendulums are defined so that, to all practical purposes, it is a single point mass vibrating. This happens when rest of the pendulums body can be considered to have effectively no weight or negligible weight compared to the "bob" or point mass[2]. Compound pendulums are so that the weight is evenly distributed along a rod and the hinge is so that there is a counterbalance effect. This means that when the hinge point is adjusted to centre of mass there is no simple harmonic motion effect[3]. Finally, the bifilar pendulum is also a rod but it is suspended by two strings and made to oscillate on the x plane and not the y plane as the others were. In this experiment the two strings can be brought closer or further apart to affect the radius of the pendulum [4].

If a pendulum is set in motion so that is swings back and forth, its motion will be periodic and this time frame for a single swing we label with T and measure in seconds. Similarly, a pendulum can be described as having a frequency which is essentially the number of swings this pendulum would produce in a minute. This value can be easily acquired using the formula T = l/f and vice versa f = 1/T

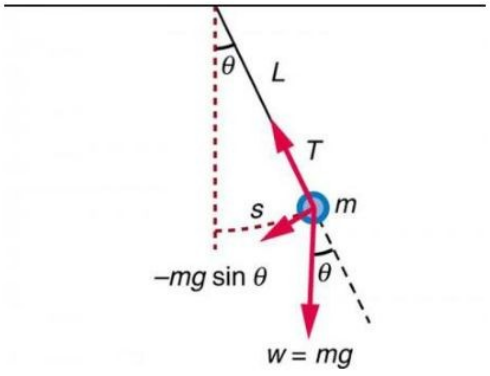
Experiment 1 Simple Pendulum

Aim

To use a simple Pendulum to measure the constant of "g"

Underlying Physics

A simple Pendulum in its simplest form can be described as a point mass suspended by a massless string from a hinge point which doesn't move or change during the swing. When a simple pendulum is displaced from its equilibrium position, there will be a restoring force that moves the pendulum back towards its equilibrium position. This restoring force is caused by its weight which is affected by gravity. [5]



As shown in the diagram the restoring force of the swinging ball I caused by mgsinθ and this can further be demonstrated using some basic trigonometry as shown:

$$sin\,\theta = \frac{0}{H}$$

$$sin\,\theta = \frac{F}{mg}$$

$$F = mg\,sin\,\theta$$

Geometrically, the arc length, s, is directly proportional to the magnitude of the central angle, θ, according to the formula s = rθ. In this diagram the radius of the circle, r, is equal to L, the length of the pendulum. Thus, s = Lθ, where θ must be measured in radians. Substituting into the equation for simple harmonic motion, we get

$$F = - ks$$

Now, if we substitute s for Lθ and the restoring force for the equation previously obtained we get

$$mg\,\sin\theta = - k(L\theta)$$

At this point we solve for the "spring constant" or k for a pendulum and it changes to

$$mg\,\sin\theta = k(L\theta)$$

$$k = \frac{m\,sin\,\theta}{L\theta}$$

Now it is important to realise that the value of sin θ approximates the value of θ for small angles in radians. Therefore, we can conclude that

$$k \approx \frac{mg}{L}$$

At this point we need to introduce the simple harmonic motion equation for the period of an oscillating system. This equation is derived from the time period of oscillation of a spring. It is independent of gravity. Instead it only depends on the mass of the system and the spring constant. The formula is

$$T = 2\pi\sqrt{\frac{m}{k}}$$

At this point all that is left is to substitute the k equation obtained previously

$$T = 2\pi\sqrt{\frac{m}{mg/L}}$$

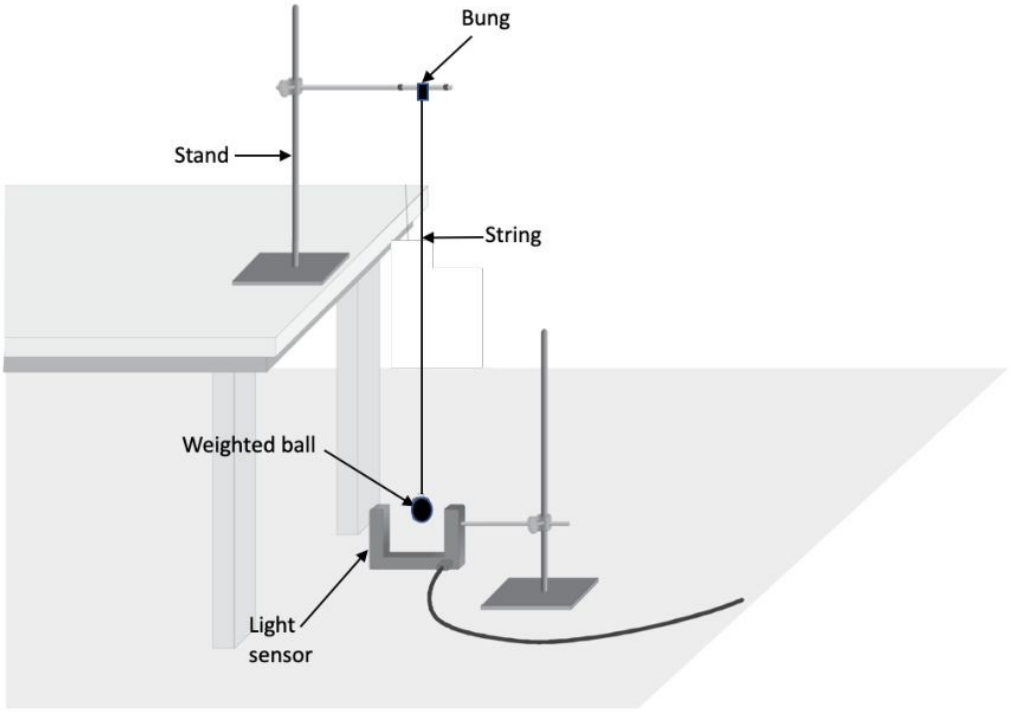
$$T = 2\pi\sqrt{\frac{L}{g}}$$

From here it's a simple rearrangement for g

$$g = 4\pi^2\left(\frac{L}{T^2}\right)$$

Apparatus
Weighted Ball
String (light)
Bung with slit for string
Clamp
Clamp stand
Pasco Capstone light sensor

Method



A weighted ball is attached to the end of a relatively light string that can easily be lengthened and shortened. The string is threaded through a bung so that the hinge point during the swing never changes. A light sensor is placed behind the ball so that the ball separates a light source from the sensor (this can also be done with a motion sensor). From this point the period of each swing can be measured at different lengths of swing. Repetitions of the found period are taken for different lengths of string so to be later averaged and squared. This information is then used to produce a graph to calculate a gradient.

Data

Experiment 1 Penduilum data						
Length (m)	Average swing times (s)					
	Trial #1	Trial #2	Trial #3	Trial #4	Trail #5	Avg of trials
0.3	11.27	11.27	11.3	11.27	11.27	11.276
0.4	12.95	13	12.95	12.98	12.98	12.972
0.5	14.4	14.4	14.35	14.4	14.35	14.38
0.6	15.75	15.75	15.75	15.75	15.75	15.75
0.7	16.95	16.95	16.95	16.95	16.975	16.955

Calculation examples

Simple Average by Adding up values and dividing by total number of data points
For example

$$\frac{11.27 + 11.27 + 11.27 + 11.3 + 11.27}{5} = 11.276$$

The values for the length of string and the squared period of the swing need to be obtained, before we can calculate the value for g using the formula:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

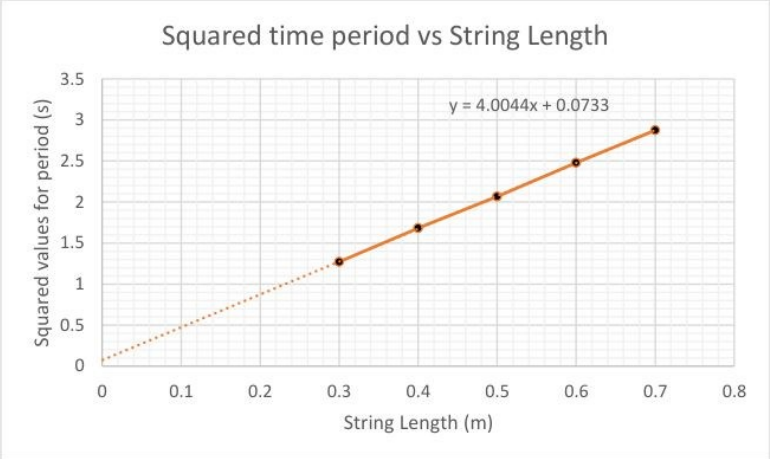
$$T^2 = 4\pi^2 \cdot \frac{L}{g}$$

$$g = 4\pi^2 \cdot \frac{L}{T^2}$$

And so, this data was calculated:

String lengt	Avg. Swin	Avg. singl	squared values	String length (m)	squared value (s)
0.3	11.276	1.1276	1.271482	0.3	1.27148176
0.4	12.972	1.2972	1.682728	0.4	1.68272784
0.5	14.38	1.438	2.067844	0.5	2.067844
0.6	15.75	1.575	2.480625	0.6	2.480625
0.7	16.955	1.6955	2.87472	0.7	2.87472025

This means that we can set a trendline of L against T squared and use that gradient to find the value for g as shown:



Therefore, with this data we can calculate the value for g:

$$m = \frac{4\pi^2}{g}$$

$$\frac{1}{4.0044} \cdot 4\pi^2 = g$$

$$g = 9.858759765$$

$$g = 9.86 ms^{-2}$$

Uncertainties

Since the g value depends entirely upon gradient then we can conclude that the gradient uncertainty accounts for the total uncertainty given the relatively tiny scalar and reading uncertainties. The gradient uncertainty was calculated as follows:

Linest function gave:

m	c
4.004374	0.073293
0.022506	0.011695

The uncertainty in the gradient is the uncertainty in g. This gave a percentage uncertainty of:

$$\%m = \frac{0.022506}{4.004374} \times 100 = 0.6\%$$

From here the percentage uncertainty of the gradient could be calculated

This uncertainty can be taken as the total uncertainty in g.

Therefore, uncertainty is
$9.86 \cdot 0.006 = 0.06$

Conclusion

In relation to the hypothesis first set the value obtained is $9.86 \pm 0.06\ ms^{-2}$

Evaluation

The results obtained by this experiment bear remarkable accuracy and precision to the given value for g. Although, perhaps a more precise measuring implement should have been used compared to a wooden rule – such as a caliper. Also, in this experiment I made no attempt to regulate the arc length of the swing- although this should make no difference to the period in theory, it may have yielded more regular results to control this aspect of the experiment so that the radian approximation makes less of a difference. Secondly, there was a systematic uncertainty which, later realized, was caused due to measuring the length of the string instead of measuring to the center of the point mass. Only measuring the string length meant the line didn't pass through the origin in the graph produce. To solve this problem, you should instead measure from the hinge of the swing to the central point mass of the weight at the end of the string. Finally, taking the average of 20 swings instead of 10 would have yielded a more reliable value for the average swing period.

On the positive side the experiment was repeated many times making the results more reliable as well as reducing any uncertainties. Additionally, the Pasco Capstone light sensor gave 20 readings every second which allowed for a precise value for the period.

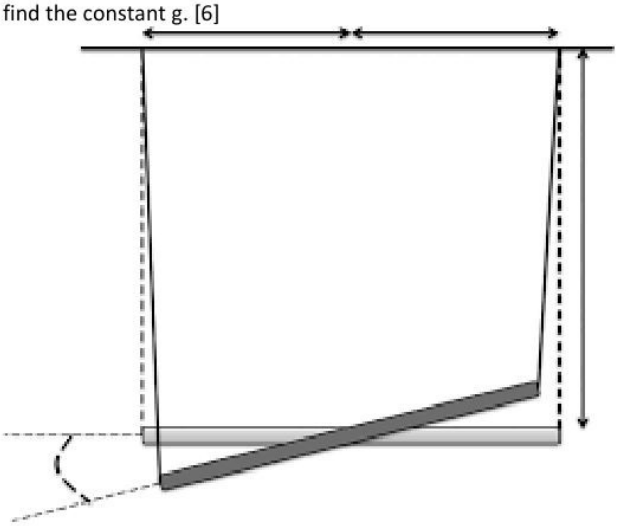
Experiment 2 – Bifilar Pendulum

Aim

To use a Bifilar Pendulum to measure the constant g

Underlying Physics

A Bifilar pendulum is defined as the suspension of an object by two wires and allowed to move freely in the x-axis. This is best displayed using a rod with wires of negligible weight compared to the rod. The classic experiment with this pendulum is used to find the rods inertia but it can also be used to find the constant g. [6]



Firstly, what is noticed about this pendulum is that the tension of the string can be seen to be the restoring force of the pendulum. It can described as follows:

$$T = \frac{mg}{2}$$

This is because the weight is spread between the two wires suspending the rod. From this point we can establish what the greater tension of the string would be when displaced from the equilibrium. This is done using basic trigonometry:

Restoring force = $T \sin\theta = \frac{mg \sin\theta}{2\cos\theta} = \frac{mg \tan\theta}{2}$

Since the restoring force acts upon both sides of the pendulum (for both strings) then the total restoring force is:

$$F = mg\ tan\,\theta$$

Since $tan\theta \approx \theta$ at small angles, then this can be further simplified. Secondly the force is made negative to describe the direction:

$$F = -mg\,\theta$$

Now, since the horizontal displacement of the can be describing with respect to the radius and theta as shown:

$s = r\theta$
Or,
$y = h\theta$

$$\theta = \frac{y}{h}$$

Where y is the horizontal displacement and h is the height of the string suspending the pendulum. Then at this point we can describe theta in this way with respect to the restoring force previously derived.

$$F = -mg\frac{y}{h}$$

This equation can now be taken and considered with torque. Since,

$$\tau = Fr$$

Then,

$$\tau = -\frac{mgyr}{h}$$

At this point we need to consider other equations simultaneously. Both:

$\tau = I\alpha$
And,
$$\alpha = \frac{a}{r}$$

Which when combined give:
$$\tau = \frac{Ia}{r}$$
And therefore,

$$\frac{Ia}{r} = -\frac{mgyr}{h}$$

$$a = -\frac{mgr^2y}{Ih}$$

For objects undergoing simple harmonic motion, as this pendulum is, then this equation hold true:

$$\alpha = -\omega^2 y$$

Where omega can be considered in terms of frequency then,

$$\alpha = -\frac{4\pi^2}{T^2} \cdot y$$

Therefore, when this is set this against the restoring force as before we. When we do this we find:

$$\frac{4\pi^2}{T^2} = \frac{mgr^2}{Ih}$$

$$\frac{T^2}{4\pi^2} = \frac{Ih}{mgr^2}$$

The moment of inertia that we would expect from a rod about its center would be:

$$I = \frac{1}{12}ml^2$$

And we can sub this into the equation to find:

$$\frac{T^2}{4\pi^2} = \frac{\frac{1}{12}ml^2h}{mgr^2}$$

$$T^2 = \frac{\pi^2 l^2 h}{3gr^2}$$

At this point we can graph T^2 $against$ $\frac{1}{r^2}$ $as\ shown$:

$$T^2 = \frac{\pi^2 l^2 h}{3g} \times \frac{1}{r^2}$$

$$y = mx$$

At this point we would notice that:

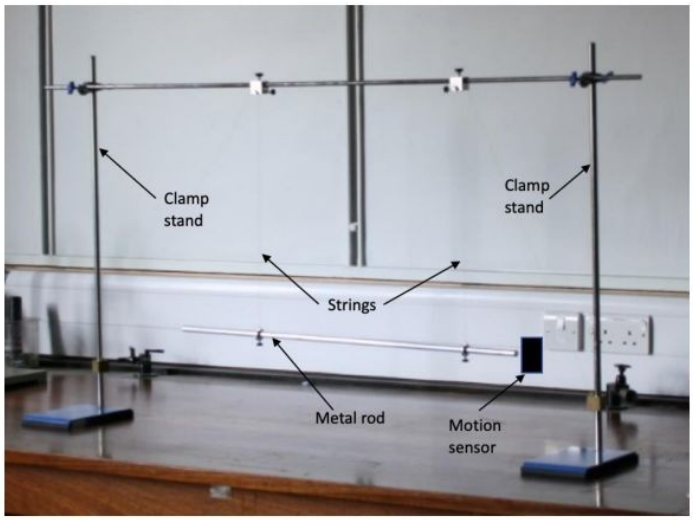
$$m = \frac{\pi^2 l^2 h}{3g}$$

And so, with a simple rearrangement we can find a value for g as shown here:

$$g = \frac{\pi^2 l^2 h}{3 \times gradient}$$

Apparatus
Light sensor
Rod (metal)
Light source (torch)
String
Long Metal Ruler

Method



A metal rod was first measured, and the midpoint found and marked. This midpoint was then used to mark lines at every 5cm intervals starting at a 10cm radius. Then the rod was hung by strings that we both tight but easily loosened and adjusted. In this way, the string was placed at the first mark on the rod at 10cm. A light or motion senor was placed at the end of the pendulum so that it could track the swings as the pendulum moved. A torch was used to make this easier.
The pendulum was then made to swing with a small arc length and the total time was taken for 10 oscillations of the pendulum. This is then repeated 5 times to obtain an average. The strings are then moved to the next position as marked on the pendulum and the whole process is repeated for all points on the pendulum.

Finally, using the information, gathered the time for one swing is calculated and then that value is squared. This result is then plotted against 1 over the radius squared so that the gradient can be used in the equation previously derived.

Data

Radius (m)		Period of Swing					Averges		
		Trial #1	Trial #2	Trial #3	Trial #4	Trial #5	Avg.	Avg. peric	Avg. period squared
0.1		34.2	34.2	34.2	34	34.2	34.16	3.416	11.669056
0.15		22.9	22.8	22.8	23	22.9	22.88	2.288	5.234944
0.2		16.8	17	16.7	16.8	16.7	16.8	1.68	2.8224
0.25		13.6	13.4	13.4	13.6	13.4	13.48	1.348	1.817104
0.3		11.2	11	11.2	11.2	11.2	11.16	1.116	1.245456

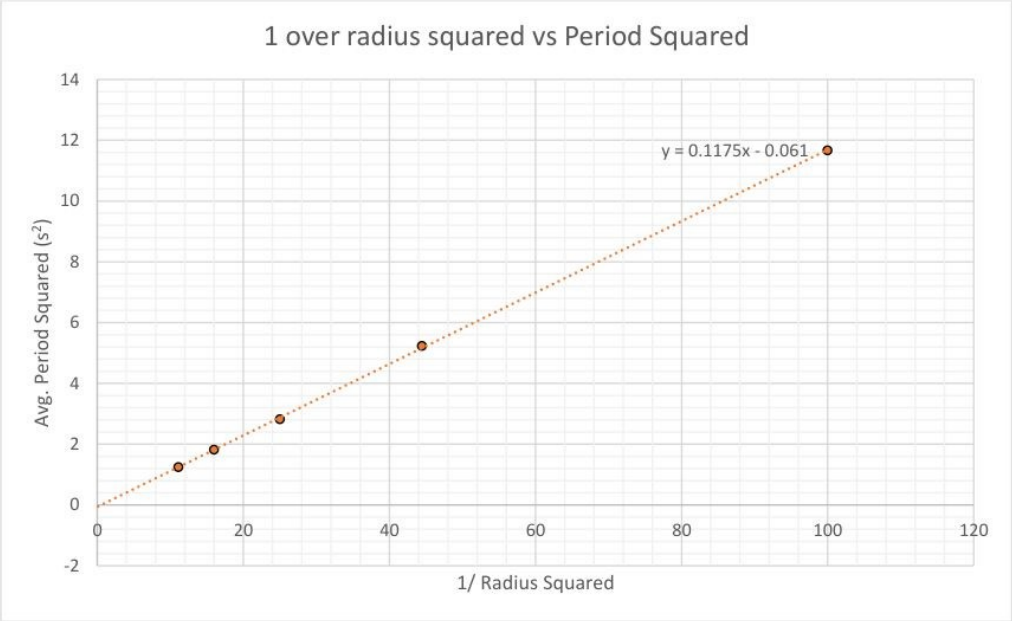
Also recorded both the height of the string suspending the rod and the length of the rod:

Length of rod (cm)	height of string (cm)
75	63.5

Using this data, I set the period squared against one over the radius as shown:

1/ Radius squared	Avg. period squared
100	11.669056
44.44444444	5.234944
25	2.8224
16	1.817104
1.11E+01	1.245456

Graph produced by data:



Calculations

Since,

$$T^2 = \frac{1}{r^2} \cdot \frac{\pi^2 l^2 h}{3g}$$

And we are producing a trendline of,

$$y = xm$$

Then,

$$g = \frac{\pi^2 l^2 h}{3 \cdot gradient}$$

$$g = \frac{\pi^2 \cdot 0.75^2 \cdot 0.635}{3 \cdot 0.1175}$$

g =10.00084914

$g = 10.00\ ms^{-2}$

Uncertainty

For this experiment the gradient uncertainty must be considered along with the reading uncertainties because they cannot be considered as part of the gradient uncertainty as they were in the previous experiments.

m	c
0.117493	-0.06097
0.000746	0.037994

From this point the uncertainty in the gradient can be calculated.

$$\%m = \frac{0.000746}{0.117493} \times 100 = 0.6\%$$

Total percentage uncertainty can be can now be calculated when assuming the reading uncertainty was 1% for both the height and length of the rod. This is done using the Pythagorean equation as follows:

%uncertainty for g

$$=\sqrt{\%gradient\ uncertainty^2 + \%length\ uncertainty^2 + \%height\ uncertainty^2}$$

$$\%\text{uncertainty for g} = \sqrt{0.6^2 + (1 \times 2)^2 + 1^2}$$

%uncertainty for g = 2.3%

This means that the total uncertainty $\pm 0.23 ms^{-2}$

Conclusion

With respect to the aim set, the value of g found using a Bifilar Pendulum was $10.00 \pm 0.23 ms^{-2}$

Evaluation

In Evaluation of the results I find them to be both accurate and precise. Although the results do not quite capture the accepted true value for g this was essentially expected given that there was a unavoidable systematic uncertainty that was caused by the rod. The rod itself was inconsistent as at one end had a notable screw fitting while the other end did not. For the purposes of this experiment I ignored this but it weighted side of the rod over the other.

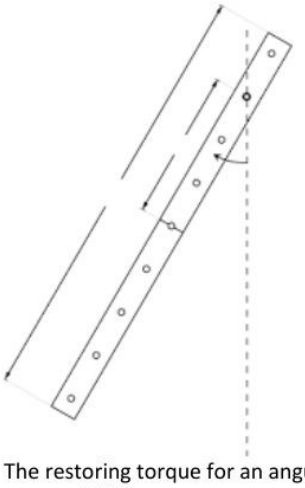
Other than this I took as many reasonable precautions to reduce the uncertainty as possible. For example, I marked on the rod at every 5-centimeter interval before placing on the pendulum- this allowed for much greater accuracy when adjusting the strings.

As before, the Pasco Capstone sensor allowed for a very high level of precision when it came to taking a times for the period because of the high rate of data points collected every second. This helped to reduce the uncertainty and increase precision greatly.
However, during the experiment I had to make several attempts to make the pendulum oscillate properly.

Experiment 3 – Compound Pendulum

Aim
Determine a value for "g" using a compound pendulum

Underlying Physics
When the weight and length of the suspended pendulum are not negligible in comparison with the distance from the axis of suspension to the center of gravity [7]. In other words, when there is a noticeable distance between; the center of gravity and the hinge point, and the hinge point to the end of the pendulum. In this case the pendulum is called a compound pendulum. A rigid body mounted so as to vibrate under the force of gravity is a compound pendulum.
Now, under these conditions, gravity acts as a torque to rotate the whole body of the pendulum. [8]



The restoring torque for an angular displacement θ is:
τ = - mg l sinθ
Where l is the distance to the centre of mass

As before in the previous experiment we can reduce this since for small amplitudes (θ ≈ 0) and so therefore:
= - mgl θ

When considering other equations for torque we can conclude that:

$$I\frac{d^2\theta}{dt^2} = -mgl\theta,$$

Where I is the inertia of the body of a pendulum

This equation represents a simple harmonic motion and so the period can be written as

$$T = 2\pi\sqrt{\frac{I}{mgl}}$$

Now, where I_G is the moment of inertia of the body about an axis parallel with axis of oscillation and passing through the centre of gravity G.

$$I_G = mK^2$$

where K is the radius of gyration which the centre of mass travels through

Using this information, we can once again rewrite the equation for the period:

$$T = 2\pi\sqrt{\frac{mK^2+ml^2}{mgl}} = 2\pi\sqrt{\frac{\frac{K^2}{l}+l}{g}}$$

Comparing this to the equation for the period of a simple pendulum, which is:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

We can conclude that

$$L = l + \frac{K^2}{l}$$

By multiplying each term by **l** and rearranging the equation we can reach a simple quadratic formula as shown:

$$l^2 - lL + K^2 = 0$$

Examining this equation, we can start to conclude certain things which we would expect about the nature of this pendulum if we were to be correct. For one, there are two roots for l which will satisfy this equation- this implies that for a given period of a swing there will be two points on the pendulum which are different distances from the end of the pendulum and yet yield the same result for the period of the swing. Since you can consider this pendulum to have effectively infinite number of values for K (because the radius of gyration can be minutely adjusted) you similarly notice that there are an infinite number of these pairings for l values.

Further, given physical necessity both values for l_1 and l_2 must be positive and we can additionally go on to conclude that:

$$l_1 + l_2 = L$$

Ultimately this means that the value for L can be obtained using the method as shown below [9]

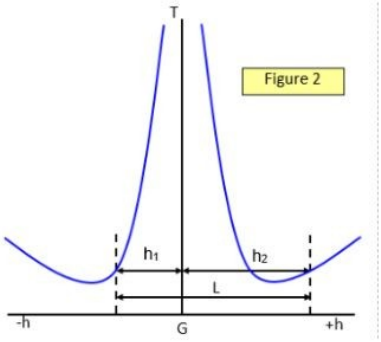


Figure 2

And it is this information that I will be obtaining in my experiment to substitute into the equation already used in the first experiment. As shown:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$g = 4\pi^2 \cdot \frac{L}{T^2}$$

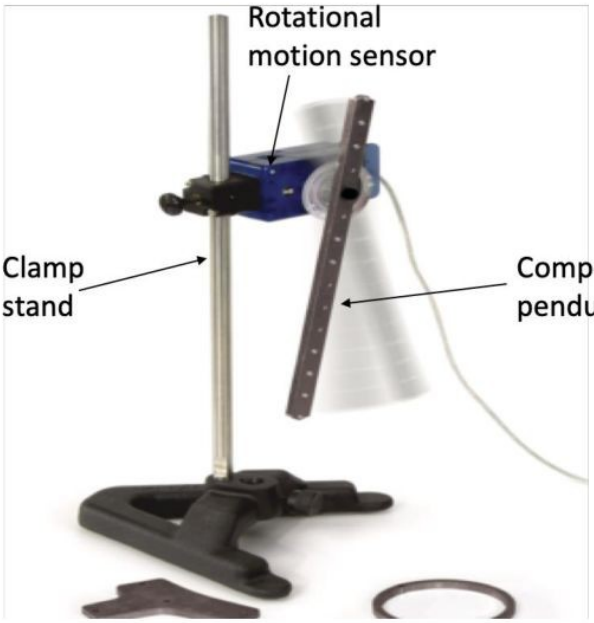
Apparatus

Pasco Physical Pendulum set

Pasco rotational motion sensor

Clamp

Clamp stand



Method

Using apparatus from a pendulum set- as shown in the image above- measurements of the distance from the multiple hinge points to the top of the pendulum were taking by a ruler (this scalar uncertainty can be virtually ignored if the pendulum was manufactured, as ours was). At this point, the pendulum is attached to a Pasco capstone rotational sensor so that the either the rotation or displacement of the pendulum can be measured over time. Measurements for the time of 20 oscillations were taken for increasing hinge points on the pendulum so that the results can be averaged and the period of a single oscillation can be calculated. By using the graph produced by this data, the value for L can be found for each period data point found in the experiment- provided there is a corresponding data on the other side of the graph.

Data

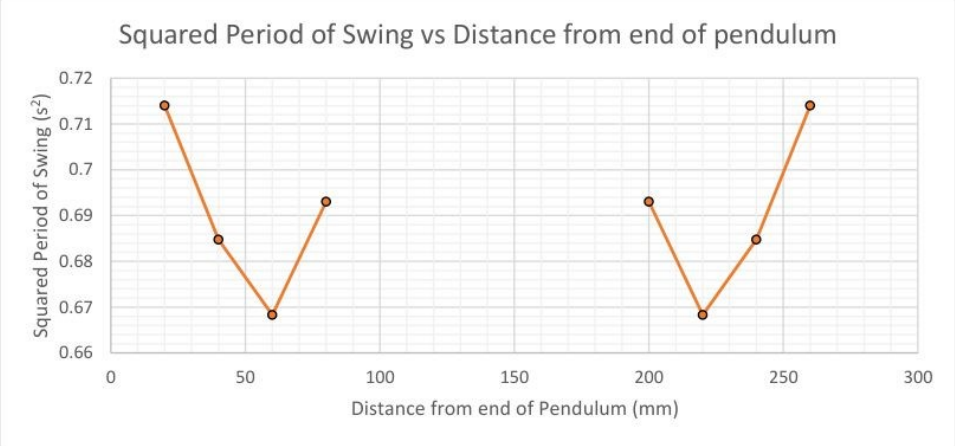
		time for 20 oscillations (s)			average swing time (s	period (s)	period squared
Serial NO	distance from end(mm)	Trial #1	trial #2	trial#3			
1	20	16.9	16.9	16.9	16.9	0.845	0.714025
2	40	16.55	16.55	16.55	16.55	0.8275	0.68475625
3	60	16.35	16.35	16.35	16.35	0.8175	0.66830625
4	80	16.65	16.65	16.65	16.65	0.8325	0.69305625
1	260	16.9	16.9	16.9	16.9	0.845	0.714025
2	240	16.55	16.55	16.55	16.55	0.8275	0.68475625
3	220	16.35	16.35	16.35	16.35	0.8175	0.66830625
4	200	16.65	16.65	16.65	16.65	0.8325	0.69305625

Calculation example

$\frac{16.9+16.9+16.9}{3} = 16.9$ $\qquad 0.845^2 = 0.714025$

$\frac{16.9}{20} = 0.845$

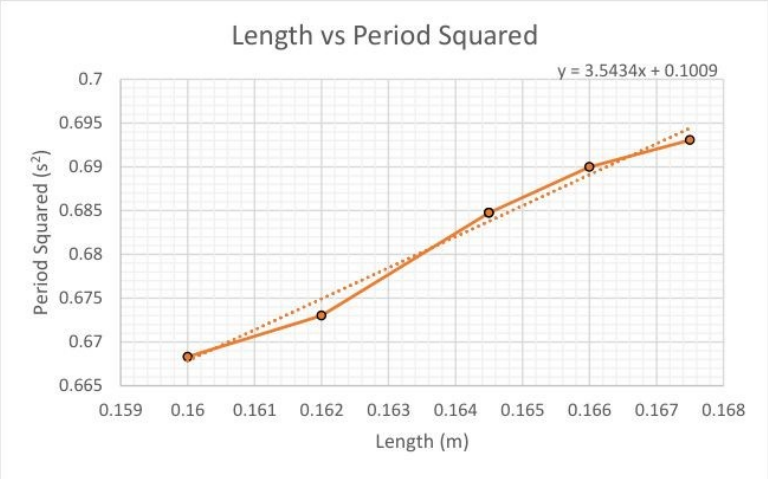
Graph



Values of L obtained by graph against T squared

L (m)	Period squared (s)
0.1675	0.69305625
0.166	0.69
0.1645	0.68475625
0.1645	0.68475625
0.162	0.673
0.16	0.66830625

Graph of L against T squared and trendline



Equations

Using the trendline given we can calculate the value of g as follows:

$$g = \frac{4\pi^2}{m}$$

$$g = \frac{4\pi^2}{3.5434}$$

$$g = 11.141394594 ms^{-2}$$

Uncertainties

While using this method it became apparent that the difference of a single millimeter on the taken value of L had a huge effect on the gradient given by the graph of T squared against L. This is because the difference between the data points along the whole graph was 7.5mm. Essentially the scaled up the reading uncertainty hugely.

The LINEST function was used to get the uncertainty in the gradient:

m	c
0.277122	-0.025
0.018786	0.012819

$$\%m = \frac{0.018786}{0.277122} \times 100 = 6.8\%$$

This gradient uncertainty must also be considered with the reading uncertainty from the graph and the scalar uncertainty from the measurements taken. All of these uncertainties can be combines using the Pythagorean algorithm as follows:

%uncertainty for g=

$$\sqrt{\%gradient\ uncertainty^2 + \%reading\ uncertainty^2 + \%scalar\ uncertainty^2}$$

$$\sqrt{6.8^2 + 5^2 + (1 \times 2)^2}$$

= ±8.7%

Therefore, the total uncertainty is found to be ±0.97

Conclusion

With respect to the aim, the calculated value for g using a compound pendulum was found to be $11.1 \pm 0.97 ms^{-2}$

Evaluation

The most pressing difficulty encounter during this experiment was the distinct lack of hinge points the were available to use. In fact, ideally, setting up a compound pendulum where the hinge point could be adjusted freely would have the potential to yield far more data points and so attain far more reliable values for L. During this experiment, I was forced to extrapolate two curves from 4 points each – this was by far the most unreliable aspect when obtaining the values for L. If more data points were taken the curve obtained would have been much more reliable.

Secondly, there was a dampening effect that was not calculated. Since the oscillations of the pendulum were seen to come to a stop roughly at the 30 swings mark, we can tell that there was some resistance in the system – this resistance will have had a small effect on the period of the swing. However, the effects of the dampening on the period are only really noticed when there is a more dramatic fall in the peak velocity of the swing. For this reason and for the purposes of this experiment, this effect was ignored.

Finally, there was a potential to take the results from two more hinge points either side of the mid hinge point (which would have yielded no results as, at any position, the pendulum would have been in an equilibrium position). Although ideally theses results would have been taken, due to time pressures this was impossible. More data points would have helped immensely at reducing any uncertainty.

More positively all points that were originally recorded were not only entirely consistent but also perfectly symmetrical. This, I believe, was due to the equipment being used. The Pasco rotational sensor allows for a single device to record the time against the oscillations. This was unlike the other experiments as they required the pendulum set up so to affect the light being received into a sensor. This experiment was respectively very easy to set up. - twenty every second - which was used to calculate the period.

Overall evaluation

As a summery, there was a very consistent accuracy of the experiment which can be attributed to the Low scalar and reading uncertainties throughout the experiment. This was ultimately due to the Pasco Capstone software which offered a very high rate of data points which was more than sufficient for these experiments. The precision of the experiments did, although, vary quite a bit - from the best uncertainty of 1.2% to the worst of over 14%. This was caused by a poor method for the available equipment – there were simple not enough hinge points on the compound pendulum that were necessary for a good reading.

If the experiments were to be redone more care would be taken to keep the arc lengths of all the pendulums both small and consistent. This is because, in every derivation implemented for the experiments, there was a small angle approximation. When measuring the values for the compound pendulum this was not appropriately taken into account and so the results may have been put off because of this.

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http://www.schoolphysics.co.uk/age16-19/Mechanics/Rotation%20of%20rigid%20bodies/text/Compound_pendulum/index.html [9]

End of script 2. **Fill in** grid 2 on your worksheet now.

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5 Script 3: Surface tension of water

This project used three procedures: drop weight, direct pull and Jaeger's method. It is 24 pages long.

Read the whole script first. Then **fill in** grid 3 on your worksheet.

Candidate 3 evidence, Physics Advanced Higher Project 2023-24, Understanding Standards. © Qualifications Scotland (SQA).

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INVESTIGATION OF SURFACE TENSION OF WATER

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1. Introduction

1.1. Abstract

This report is to compare different methods of finding the surface tension of water as well as to determine the relationship between temperature and the surface tension of water.

After the experiments it was determined that the direct pull method this is because it produced values very similar to the values expected under the circumstances. The average value collected in the experiment being $0.0727 Nm^{-1}$ compared to the value of $0.0724 Nm^{-1}$. [4]

It was found from the graph for the jaeger's method experiment that the surface tension of water was directly proportional to temperature, as temperature increases the surface tension of water also increases with the equation being y=0.0103x-0.1679 this is the equation we calculated which was a far bit off the actual relationship between temperature and surface tension.

1.2. Background to surface tension

The strong attractive forces between liquid molecules are the reason that we have surface tension. The molecules on the surface do not have other molecules like them on the surface this makes them have a strong coherent force and attract to each other more strongly on the surface. With this coherent force it creates a kind of 'film' on the surface of the water, this film makes it harder to move an object through the surface than if the object is completely submersed.

Surface tension is measured in Nm^{-1} (force per unit area), the force in newtons required to break a film of 1m length. Different liquids have different surface tensions with water having the surface tension of $0.0724 Nm^{-1}$ [4] compared to liquids ethyl alcohol and mercury being $0.0223 Nm^{-1}$ and $0.465 Nm^{-1}$ [4].

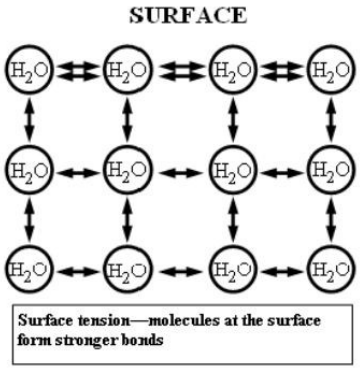


Figure 1.2- Interactions between water molecules.

This diagram show how surface tension makes the "skin" on top of the water which is just a stronger force of attraction between the molecules on the surface compared to the rest of the water. With the arrows symbolising the attracting forces between the water molecules.

2. Experiment 1 – drop weight method

2.1. Underlying physics

In the drop weight method, a liquid is introduced in large volumes through a capillary. A drop, which tries to move upward due to buoyancy, forms at the tips of the capillary. Because of this, the interfacial tension y the drop tries to keep as little contact with the bulk volume travelling through the capillary. As a new comes into being when the drop leaves the end of the capillary tube, it is necessary for the drop to overcome the corresponding interfacial tension. The drop wants to have as little contact with the bulk as possible with a new interface appearing when the initial drop, drops from the capillary tube. With this the drop will need to over come the corresponding interfacial tension. The drop will not drop until the weight or the lifting/buoyancy force compensates the interfacial force. [5]

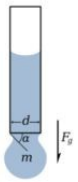


Figure 2.1- combinations of forces acting on a drop exiting a capillary tube

The force from the internal pressure must first be calculated. The internal pressure is given by $\frac{y}{r}$. [5]

$$F = PA$$

$$= \frac{y}{r} \times \pi r^2$$

$$= y\pi r$$

Then the sum of the downward forces is:

$$F = y\pi r + \mathrm{mg}$$

The upwards force is given by using the formula for surface tension,

$$y = \frac{Force}{Length}$$

So

$$F = yl$$

Force this experiment l is the circumference of the drop so $2\pi yr$,

$$F = 2y\pi r$$

By then equating the upwards and downwards forces we can find the equation for surface tension, [2].

$$2\pi ry = y\pi r + \mathrm{mg}$$

$$y(2\pi r - \pi r) = \mathrm{mg}$$

$$y(\pi r) = mg$$

$$y = \frac{mg}{\pi r}$$

2.2. Procedure

All the glassware was scrubbed clean and soaked in an acid bath. A clamp stand was then set up holding a burette, attached to this burette was an extra glass tube this was to reduce the flow of water with a clamp being on the connector to get only small drops out of the apparatus. The beaker was then placed under the apparatus after it was weighed on a balance. The beaker was always emptied and dried, so the mass was the same every time. Then water was placed in the burette and the flow was severely restricted due to the clamp to one drop per minute. This time intervals allows the drop to form correctly. A counted number of drops (10) were collected in the beaker. The total mass was found was found and the mean mass of a drop was also found. To measure the radius of the tube a set of callipers were used.

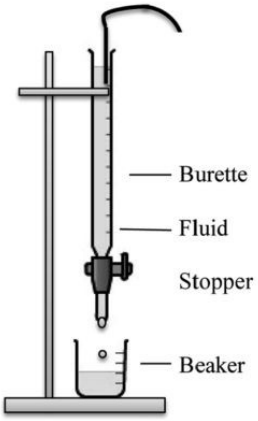


Figure 2.2- Functional diagram of drop weight.

This diagram doesn't include a stop watch that was used to time the intervals of the drops and the callipers which was used to measure the inner radius of capillary tube.

2.3. Results

Mass of Beaker (g)	**Mass of beaker + drops**	**Number of Drops**
39.43	40.15	10
39.42	40.18	10
39.4	40.25	10
39.41	40.24	10
39.41	40.23	10
Mean mass of Drops (g)	**Radius of orifice (mm)**	**Surface tension (Nm^{-1})**
0.072	3.6	0.0623
0.076	3.6	0.0659
0.085	3.6	0.0737
0.083	3.6	0.0719
0.082	3.6	0.0711
Overall mean mass of drop (g)	**Average surface tension (Nm^{-1})**	
0.080	0.069	

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2.4. Analysis

The formula used for calculating the surface tension:

$$y = \frac{\text{mg}}{\pi r}$$

g- acceleration due to gravity which is 9.8ms^{-2}

r- the radius of the orifice and it was measured to be 3.6mm

m- mass of a singular drop

$$mass\ of\ drop = \frac{Mass\ of\ beaker\ and\ drops\ -\ mass\ of\ beaker}{10}$$

$$= \frac{40.15 - 39.43}{10}$$

$$= 0.072 \times 10^{-3} kg$$

The mean mass of the 5 experiments was then found,

$$\bar{m} = \frac{0.072 + 0.076 + 0.085 + 0.083 + 0.082}{5}$$

$$= 0.080 \times 10^{-3} kg$$

These values were then subbed into the equation:

$$y = \frac{\text{mg}}{\pi r}$$

$$= \frac{0.08 \times 10^{-3} \times 9.8}{\pi \times 3 \cdot 6}$$

$$= 0.069 Nm^{-1}$$

2.5. Uncertainties

Errors occurred in the measurements of the mass of each of the drops, m, and in the radius of the tube r.

For m, the random uncertainty and the scale reading uncertainty must be found.

$RandomUnc = \frac{(Max - min)}{n}$

For the example for m:

$RandomUnc = \frac{(0.85 - 0.72)}{5}$

$RandomUnc = \pm 0.026g$

Scale reading uncertainty is half the smallest unit on the scale which is $\pm 0.01g$

To combine the uncertainties, we must convert them into percentage form

$\%unc = \frac{Unc}{Value} \times 100$

For example

$$\frac{0 \cdot 01}{0.80} \times 100 = \pm 1.25\%$$

The combination follows the following formula

$$Unc^2 = (\%unc)^2 + (\%unc)^2 + \ldots$$

$$Unc^2 = ((3.25)^2 + (1.25)^2)$$

$$Unc = \pm 3.48\%$$

The uncertainty for one drop is

$$= \pm 0.348\%$$

We must combine the uncertainty of r and the uncertainty for one drop

$$Unc^2 = (0.348)^2 + (0.277)^2$$

$$Unc = \pm 0.44\%$$

Absolute uncertainty $= \frac{0 \cdot 44}{100} \times 0.069$

$= \pm 0.0003 Nm^{-1}$

2.6. Evaluation

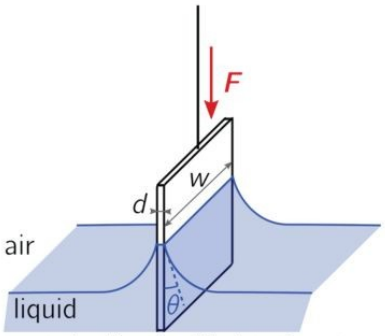
Comparing the results that we have gained from this experiment for surface tension ($0.069 Nm^{-1}$) with the actual value of surface tension ($0.0742 Nm^{-1}$) we can see that both values have the same magnitude. The experimental values came out to be smaller than the actual value, this is due to the uncertainties of a possibility of a mark on the glass or a draft making the drop detach prematurely leading to an uncertainty in the mass of the drop. We could sort this out by putting the apparatus in a draft less room.

A way that we could improve the experiment is by not emptying the beaker at the end of each experiment, with this we can can take the final mass of the previous experiment as the start mass of the next experiment. This would save a lot of time and help reduce any chance of residue water from the previous experiment.

3. Experiment 2 – direct pull method

3.1. Underlying physics

The direct pull method describes the use of a torsion balance to measure the surface tension forces acting on a microscope slide or a frame which is suspended to touch the surface of the liquid. The torsion balance is used to get a scale for the tension force acting on the slide this is done by a skin forming around the bottom of the slide and the scale reads the force needed to break the skin. This occurs as the water wants the reduce its surface area but is attracted to the slide, so a force is



created pulling the slide into the water.

Figure 3.1- forces acting upon a microscope slide

The downward force on the slide can be represented as y(2L+2t), where y is the surface tension, L is the length if the slide and t is the thickness of the slide. Using the balance beam the force acting on the slide can be counteracted by adding masses to the dish on the opposite end of the torsion balance. As a result, y(2L+2t) can be equal to the weight of the masses

Hence y(2L+2t)=mg [3]

This can then be rearranged for y to be the subject:

$$y = \frac{mg}{2t+2t}$$

This is the equation needed to find surface tension.

3.2. Procedure

A torsion balance was set up with the scale set at 0, this was done by adding masses to the opposite end of the scale to make it 0. A microscope slide and a dish was then attached to the side with the scale. A large beaker was then filled with water and placed on a lab jack. Once on the lab jack the beaker was raised to that the microscope slide just touches the water. Then the beaker is lower slowly as to be able to see when the slide breaks away from the water, just before it breaks take a note of the scale and where the indicator lies. Then the beaker is removed and then masses are added to the dish by the scale the get the same reading that was produced by the water this is the value for m in kg. The mass of the masses was calculated using a set of scales. This experiment was repeated five times with four different slides, which had their length and breadth measured by callipers. All the glassware was cleaned thoroughly.

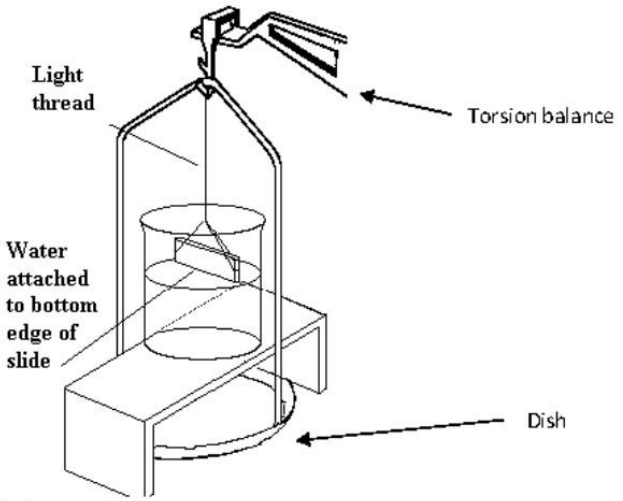


Figure 3.2- a functional diagram of direct pull method

This diagram does not include a set of callipers which was used to measure the length and breadth of the microscope slide.

3.3. Results

L (m)	B (m)	Slide	Mass (kg)					Mean mass (kg)
			1	2	3	4	5	
0.07496	0.00098	1	0.00112	0.00112	0.00122	0.00109	0.00110	0.00113
0.07532	0.00097	2	0.00110	0.00111	0.00111	0.00109	0.00108	0.00110
0.07539	0.00098	3	0.00110	0.00109	0.00108	0.00107	0.00108	0.00108
0.07507	0.00099	4	0.00120	0.00121	0.00120	0.00123	0.00120	0.00121

Since the slides were from the same manufactured batch I made a mean of the length and breadth.

L= 0.07519m

t= 0.00098m

3.4. Analysis

Using the formula to calculate surface tension:

$$y = \frac{\text{mg}}{2L + 2t}$$

g- gravitational constant due to gravity

m- mass used to get the value on the torsion balance (kg)

L- length of slide (m)

t- breadth of slide

The values from the table was then substituted into the equation

Slide 1:

$$y = \frac{\text{mg}}{2L + 2t}$$

$$y = \frac{0.00113 \times 9.8}{(2 \times 0.07519) + (2 \times 0.00098)}$$

$y = 0.0727 \text{Nm}^{-1}$

Slide 2:

$y = \frac{0.0011 \times 9.8}{(2 \times 0.07519) + (2 \times 0.00098)}$

$y = 0.07076 \text{Nm}^{-1}$

Slide 3:

$y = \frac{(0.00108 \times 9{\cdot}8)}{(2 \times 0.07519) + (2 \times 0.00098)}$

$y = 0.06948 \text{Nm}^{-1}$

Slide 4:

$y = \frac{0.00121 \times 9{\cdot}8}{(2 \times 0.07519) + (2 \times 0.00098)}$

$y = 0.07784 \text{Nm}^{-1}$

3.5. Uncertainties

With using this method uncertainties arose creating many varying results, one of which was the masses which was set on the dish the other being the length and breadth of the microscope slide. To get the uncertainty for the mass, the masses were measured multiple times to get the correct value on a balance.

The random uncertainty was calculated for the four different slides:

$$RandomUnc = \frac{Max - min}{n}$$

For example:

$$RandomUnc = \frac{1.22\times10^{-3} - 1.09\times10^{-3}}{5}$$

$RandomUnc = \pm 26\times10^{-3}$ kg

This was then converted into a percentage and combined with the length and breadth uncertainty.

$$\%Unc = \frac{unc}{Value} \times 100$$

$$\%Unc = \frac{26\times10^{-6}}{1{\cdot}155\times10^{-3}} \times 100$$

$\%Unc = \pm 2.25\%$

The random uncertainty in L was found to be $\pm 1.075\times10^{-4}$ m as

$$RandomUnc = \frac{75.39\times10^{-3} - 74.96\times10^{-3}}{4}$$

$RandomUnc = \pm 1.075\times10^{-4}$ m

The random uncertainty for t was found to be $\pm 5\times10^{-6}$ as

$$RandomUnc = \frac{0.99\times10^{-3} - 0.97\times10^{-3}}{4}$$

$RandomUnc = \pm 5\times10^{-6}$ m

The percentage uncertainty was then calculated for L and t being ±0.14% and ±0.51% respectively. With the percentage for the masses being larger than three multiples the values for L and t can be ignored.

The random uncertainty for the surface tension through the different slides was found to be $\pm 2.09\times10^{-3}$ Nm^{-1} which in percentage form is ±2.87% this will then be combined with the value for mass which comes out at ±3.65%

The definitive answer comes out at $0.0727 \pm 0.00265 Nm^{-1}$

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3.6. Evaluation

The results obtained from this experiment, when compared to the true value with the uncertainty covering the true value.

Ways this experiment could be improved is by having the glassware more thoroughly cleaned as a single mark on the glassware can cause a large uncertainty to occur. Another problem that occurred was when the indicator had to be zero on the scale on the torsion balance this caused a problem when trying to get the correct scale inclinations. Another problem which occurred was with the slide staying attached to the water with slight vibrations due to people walking by, creating oscillations in the equipment which then breaks the skin on top of the water.

4. Experiment 3 – Jaeger's method

4.1. Underlying physics

The coefficient of surface tension decreases with increasing temperature, with this water can drip into a large flask with an airtight seal, so forcing bubbles of air out of the capillary tube which drips into a beaker of water. The lower end of the capillary tube is a depth h_1 below the water surface. It can be shown that the bubble will break free from the end of the tube when the radius is equal to the internal radius of the tube. Using the manometer, the total pressure within the apparatus may be found; with this the internal pressure will be equal to the hydrostatic pressure which is ($h_1 r_1 g$). The excess pressure within the air bubble is due to surface tension of water. With the total pressure being given with the equation

$h_2 r_2 g = h_1 r_1 g + \frac{2T}{r}$ [1]

With r_1 being the density at a specific temperature, r_2 is the density of the liquid in the manometer, r is the radius of the capillary tube and with h_2 being the difference in levels within the manometer. The coefficient of surface tension of water T, with heating the water in the large flask giving a range of values for T. This equation can then be changed to get T as the subject:

$$T = \frac{1}{2} r g (h_2 r_2 - h_1 r_1)$$

4.2. Procedure

The apparatus used is show in figure 4.2. The glassware was first cleaned with and acid bath and scrubbed thoroughly to get any marks or stains off the glass that will affect the experiment. Then that glassware was assembled with there being a conical flask, a U tube, a T tube, 2 other tubes to connect them and a funnel or something else that allows water to drip slowly. Water was then added to the beaker, as well as the liquid being added to the manometer of which it is toluene (safety precautions must be made as toluene is an irritant). Once an airtight seal is in place run water through the funnel until a bubble is just about to leave the end of the tube this gets a displacement in the manometer, this displacement was then measured using a ruler being the values for h_2. The ruler was also used to find the vale for h_1 with it being the distance under the water the end of the bubble producing tube. A pair of callipers was used to find the value for r. This water done for

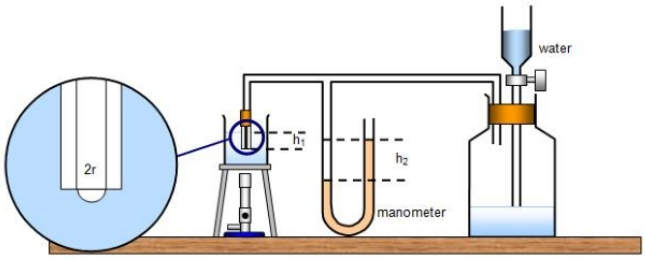


Figure 1

beaker temperatures of 22, 27, 32 and 37oC, it was also repeated four times.

Figure 4.2- a function diagram for Jaeger's method.

This diagram does not include the ruler stick which was used to measure the displacement in the manometers.

4.3. Results

Temperature (oC)	Mean displacement (m)	Density of water (kg/cm^3)	Surface tension (Nm^{-1})	Absolute uncertainty in surface tension (Nm^{-1})	Actual value of surface tension (Nm^{-1})
22	0.0295	1000	0.0467	0.0027	0.0724
27	0.0413	996.5	0.1327	0.0036	0.0720
32	0.0443	995.1	0.1547	0.0145	0.0711
37	0.0520	993.3	0.2113	0.0073	0.0706

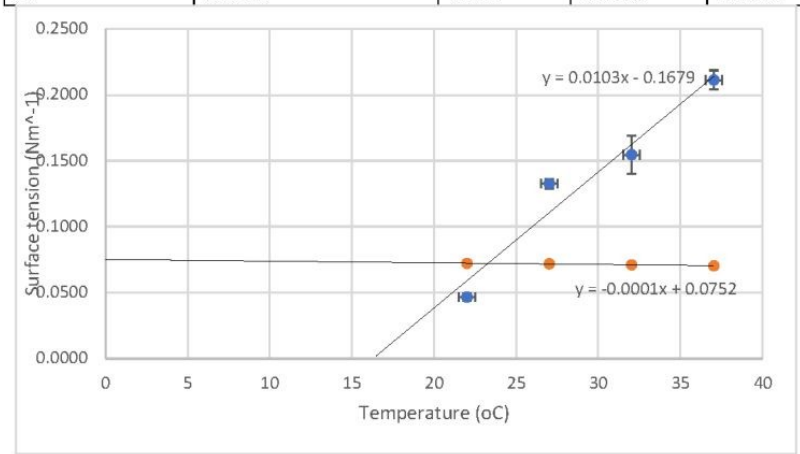


Figure 4.4- graph of results top line was calculated, and bottom line is actual values.

4.4. Analysis

The formula used to find the surface tension through jaeger's is:

$$y = \frac{1}{2} rg(h2r2 - h1r1)$$

In each experiment g is constant at $9.8ms^{-2}$, also the value for r was constant through out the experiment being 0.00171m for the radius of the capillary tube.

r2 and h1 also remained constant through out the experiment with r2 being the density of toluene which was calculated as follows

$$r2 = \frac{m}{V} = \frac{867}{1000} = 0.867kgl^{-1}$$

h1 was found by using a ruler stick and measuring the distance in the water capillary tube was this was kept constant by keeping the equipment set up and not moving any parts.

r1 was found using a table of values [6].

h2 was found out by using a ruler stick and measuring the displacement just before a bubble leave the end of a capillary tube.

The values were substituted into the equation;

$$y = \frac{1}{2} rg(h2r2 - h1r1)$$

$$y = \frac{1}{2} \times 0.00171 \times 9.8(0.0295 \times 867 - 0.02 \times 1000)$$

$$y = 0.04567Nm^{-1}$$

4.5. Uncertainties

Many uncertainties contributed to the experiment with the main one being random uncertainty of the surface tension of water which was caused by the measuring of the manometers.

$$randomUnc = \frac{max - \min}{n}$$

Example;

$$= \frac{0.031 - 0.028}{4}$$

$$= 0.00075m$$

This turned into a percentage

$$\%Unc = \frac{0.00075}{0.0295}$$

$$\%Unc = 2.5\%$$

This then onto the value for surface tension gives the absolute value once converted

$$absolute\ uncertainty = 0.0027Nm^{-1}$$

This is like in the table

4.6. Evaluation

Many problems occurred during this experiment that could have caused such a large uncertainty and completely wrong equation for the graph.

One of these reasons one of these could be because the glass wear wasn't properly clean, a way to sort this is by scrubbing it thoroughly and bathing it in acid.

another reason could be because a proper airtight seal could not be achieved, a way this could be improved is by making sure the pieces properly fit and have spares if any break.

Another reason could be because a bubble appeared in the U bend tube which can cause problems to the pressure in the apparatus.

another reason could be put down to the way it was measured using a ruler stick instead of a travelling microscope which is a more accurate form of measuring.

As can been seen from the graph the results are showing the wrong gradient for the graph as the gradient is supposed to be negative this could have been caused by the bubble occurring in the U tube and a failure to measure properly.

5. Discussion

5.1. Conclusion

The best method for measuring surface tension was found out by carrying out 3 different experiment and evaluating the results gained from them.

To determine the best method for finding the surface tension we must consider multiple factors these being; the values closest to the actual value, the values of uncertainties and the simplicity of the apparatus and how easy it was to set up and use.

The direct pull method got the closest to that actual value of the surface tension. We calculated that the surface tension through the direct pull method was $0.0727 Nm^{-1}$ this is then compared with the actual value for surface tension being $0.0724 Nm^{-1}$ with this we can see that both values are in the same magnitude with our value being out by 0.41% this is incredibly close to the actual value more so than the drop weight with it being out by 4.7% which is still close but the direct pull method still beats it out

The direct pull method was also the easiest to set up with it having a very little amount of apparatus and it being very easy to set up. Reasons for this is because of the simplicity of the method with the beaker and scale being the only variables which change in the experiment. This is unlike Jaeger's method which the experiment relies on many aspects being right for the experiment to work, for example the apparatus has to have an air tight seal as well as the water has to run through at a certain volume for the bubble to be produced, also with the pressure was affect with small movements of the apparatus which affected the results.

The most consistent results were from the drop weight method with the uncertainty for it being $\pm 0.44\%$ compared to the uncertainties for the direct pull method being $\pm 3.45\%$. This show that the drop weight method was far more consistent than the drop weight.

5.2. Evaluation

Whilst doing the experiments there were many factors which contributed to the uncertainties for example in the direct pull method it was found that the scale was sometimes offset this was caused by the counter balance weights slipping and putting off the scale a slight bit. A way this could be improved is by having a mass that makes the scale at 0 and securing it properly. This will reduce the uncertainty in the scale.

The largest uncertainty was increasing the temperature in the Jaeger's method we used a hot plate to heat the beaker in which it was hard to get proper intervals and assumptions had to be made for increasing temperature. A way this could be improved upon is by doing the experiment more times at the temperatures and by measure the temperature of the water more times before starting the experiment.

The biggest problem in the three experiments was the jaegers method as the graph shows a completely different relationship to the actual relationship this causes huge problems to its uncertainties as well.

6. References

[1] Article title: schoolphysics ::Welcome::

Website title: Schoolphysics.co.uk

URL: http://www.schoolphysics.co.uk/age16-19/properties%20of%20matter/surface%20tension/text/surface_tension_and_temperature/index.html

Date visited: 18 December 2019

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Website tittle: Kruss-scientific.com

URL: http://www.kruss-scientific.com/services/education-theory/glossary/drop-volume-tensiometer/

Date visited: 8 January 2020

[3] Author: E Armatige

Year published: 1983

Book title: Practical physics in SI

City: London

Publisher: Murray

Pages used: p30-31

[4] article title: Surface Tension

Website title: Hyperphysics.phy-astr.gsu.edu

URL: http://www.hyperphysics.phy-astr.gsu.edu/hbase/surten.html

Date visited: 15 February 2020

[5] Author: F Tyler

Year published: 1970

Book title: a laboratory manual of physics SI units

City: London

Publisher: Edward Arnold

Pages used: p30,31,32,33,46 and 47

[6] Article title: Density of Water – The Physics Factbook

Website title: Hypertextbook.com

URL: https://hypertextbook.com/facts/2007/AllenMa.shtml

Date visited: 12 December 2019

Word count: 3662

End of script 3. **Fill in** grid 3 on your worksheet now.

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6 Answers

These are the marker's marks, from the Understanding Standards commentary. **Mark the scripts yourself before you look.**

Item	Section	Max	Script 1	Script 2	Script 3
1	Abstract	1	1	1	0
2	Underlying physics	4	1	3	2
3a	Diagrams and apparatus	2	1	2	1
3b	How the apparatus was used	2	1	0	1
3c	Complexity and demand	3	3	3	2
4a	Data	1	1	1	0
4b	Analysis	4	1	3	1
4c	Uncertainties	3	1	2	0
5a	Conclusion	1	1	1	0
5b	Evaluation of procedures	3	1	2	1
5c	Overall evaluation	3	1	1	1
5d	Quality project	1	0	1	0
6a	Structure	1	1	1	1
6b	References	1	1	0	1
	Total	**30**	**15**	**21**	**11**

Focus task answers

Task 1. Script 1's graphs are small and have no minor gridlines, and the graphs on pages 13 and 17 are not appropriate for finding wavelength (4b: 1 out of 4). Script 2 analyses all three procedures by graph and uses each gradient for g (4b: 3 out of 4). Its procedure 1 graph is a little small, and procedure 3 joins the dots.

Task 2. Script 2 interprets its results, discusses problems and suggests improvements. It does not discuss the selection of procedures and suggests no further work (5c: 1 out of 3).

Task 3. Calibration, scale reading and random. Script 2 uses LINEST for the uncertainty in the gradient. Procedure 1 has no reading, random or calibration uncertainties. Procedure 2 combines the LINEST gradient uncertainty with the length and height uncertainties using the power rule (4c: 2 out of 3).

Task 4. Procedure 1 is not in the past, impersonal voice, and gives no repeats, range or interval. Procedure 2 gives range, interval and repeats, but mixes voices (3b: 0 out of 2). Script 3 uses an incorrect tense several times, so 1 mark at most.

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Task 5. There is no raw data for procedure 3, only mean values, so the surface tension values cannot be checked. 4a scored 0. The uncertainties for procedure 3 cannot be checked either (4c scored 0).
Task 6. Script 1 states the aim and a finding for each procedure (1 mark). Script 3 leaves out one method and its finding, and claims direct proportion that its graph does not support (0 marks).

The full commentary follows. Read the marker's reasons for each section.

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Commentary on candidate evidence

Candidate 1

1 Abstract

The candidate was awarded **1 out of 1 mark** because the candidate has clearly stated the aim of the project and given findings for each procedure used. The findings are consistent with stated conclusions. The abstract immediately follows the contents page.

2 Underlying physics

The candidate was awarded **1 out of 4 marks** because the candidate has included some short underlying physics paragraphs which contain some relevant background physics. There are however some mistakes:

- On page 4, there is an incorrect statement of p = velocity and confusion with m and n.
- On page 5, sinθ is not correctly defined and the rearrangement of relationship is incorrect.
- On page 8, the attempt at a derivation for lambda includes an arithmetic error.
- No further underlying physics is included for procedures two or three.

3 Procedures

The candidate was awarded **5 out of 7 marks**. The marks were awarded as follows:

3(a) **1 out of 2 marks** was awarded because the candidate has included images for all three procedures.

Most of the images are labelled correctly however replication would be difficult given the information provided.

A circuit diagram for procedure one would be desirable.

3(b) **1 out of 2 marks** was awarded because the candidate has indicated repetition of measurements in all three procedures.

The description of two of the procedures did not include the range and interval of the independent variables, and so do not support straightforward replication. The descriptions of procedures were written in appropriate tense and voice.

3(c) **3 out of 3 marks** was awarded because the candidate has completed procedures which are all at an appropriate level and original to the candidate.

With a reasonable range of independent variables and a number of repeats, the procedures would have taken 10-15 hours of lab time.

The limitations in the analysis of the data in procedure one is considered in the analysis section.

4 Results (including uncertainties)

The candidate was awarded **3 out of 8 marks**. The marks were awarded as follows:

4(a) **1 out of 1 mark** was awarded because the candidate has included data that are both relevant and sufficient for each procedure.

4(b) **1 out of 4 marks** was awarded because the table headings, units and mean values are mostly correct.

The candidate has attempted graphical analysis for two of the three procedures. However, there are a number of mistakes, for example:

- On page 8, it is not clear what the values in the table are, an incorrect unit for $V^{-1/2}$ has been given, the average values for 3kV & 5kV are incorrect, it is not clear what values the candidate is using to calculate lambda, and the final calculation is incorrect for the values given.
- On page 13, an inappropriate graph is used to determine wavelength.
- On page 17, an inappropriate graph is used to determine wavelength.
- Both graphs are small and lack minor gridlines.

4(c) **1 out of 3 marks** was awarded because the candidate has shown an awareness of random, scale reading and calibration uncertainties.

The calculated random uncertainties are correct.

Error bars are included on the graphs, but it is not clear how they were established.

No attempt has been made at determining the uncertainty in the gradient.

No attempt has been made to combine uncertainties.

5 Discussion (conclusion(s) and evaluation)

The candidate was awarded **3 out of 8 marks**. The marks were awarded as follows:

5(a) **1 out of 1 mark** was awarded because all conclusions are stated correctly.

5(b) **1 out of 3 marks** was awarded because the candidate has attempted to address limitations of equipment, adequacy of repeated readings and control of variables.

However, there are many irrelevant statements leading to a lack of quality in their evaluation.

5(c) **1 out of 3 marks** was awarded because the candidate has attempted to compare results with expected values.

The candidate has made one suggestion for an improvement to procedure three.

5(d) **0 out of 1 mark** was awarded because this is not a quality project.

6 Presentation

The candidate was awarded **2 out of 2 marks**. The marks were awarded as follows:

6(a) **1 out of 1 mark** was awarded because a contents page and page numbers are included.

6(b) **1 out of 1 mark** was awarded because three references are given in Vancouver style and are correctly cited.

Overall

The candidate was awarded a total of **15 out of 30 marks**.

Candidate 2

1 Abstract

The candidate was awarded **1 out of 1 mark** because the candidate has clearly stated the aim and findings of the project.

2 Underlying physics

The candidate was awarded **3 out of 4 marks** because the candidate has included a short introductory underlying physics paragraph about the various pendula used.

The candidate has defined most of the variables used.

Overall, the candidate has worked through the theory methodically, arriving at the relationships required to obtain g graphically, although procedure two is a little difficult to follow and there are some large algebraic jumps in procedure three.

3 Procedures

The candidate was awarded **5 out of 7 marks**. The marks were awarded as follows:

3(a) **2 out of 2 marks** was awarded because the candidate has included labelled images for each procedure.

3(b) **0 out of 2 marks** was awarded because for procedure one the candidate has not used the past, impersonal voice in their description. The candidate has not given information about the number of repetitions and the range and interval of the independent variable.

For procedure two, the candidate has given information about range, interval, and repetitions. The candidate has used a mixture of past, impersonal and present voice.

For procedure three, the candidate has not given information about number of repetitions and the range and interval of the independent variable.

3(c) **3 out of 3 marks** was awarded because the candidate has chosen three procedures all at an appropriate level for Advanced Higher Physics. The experimental procedures would have taken at least 10-15 hours.

4 Results (including uncertainties)

The candidate was awarded **6 out of 8 marks**. The marks were awarded as follows:

4(a) **1 out of 1 mark** was awarded because the candidate has included sufficient and relevant data.

4(b) **3 out of 4 marks** was awarded because the candidate has analysed their data using appropriate graphical methods for all three procedures, generating a line of best fit and determining the gradient.

The candidate's graph for procedure one is a little small. They have used the graph gradient appropriately to determine a value for g.

The procedure two graph is an appropriate size. The candidate has used the graph gradient appropriately to determine a value for g.

For procedure three, the candidate has generated the curves by joining the dots. Also, four points is insufficient to lead to a valid value of the pendulum effective length l required to determine g. It is also unclear how the value of l was derived from these graphs. This value is then used appropriately to generate a line of best fit from which a value of g was determined. The candidate has also included an inappropriate 'dot-to-dot' line.

4(c) **2 out of 3 marks** was awarded because for procedure one the candidate has used the LINEST function to correctly determine the gradient uncertainty and apply this to their value for g. The candidate has not included individual reading, random or calibration uncertainties for their data.

For procedure two, they have correctly combined their LINEST gradient uncertainty with 2xl and the height uncertainty. Both reading uncertainties are quoted as 1%. The candidate has correctly used the uncertainty power rule. They have applied this correctly to determine an absolute uncertainty for g.

For procedure three, the candidate has correctly determined the gradient uncertainty using the LINEST function. They have however, incorrectly combined reading uncertainties with the gradient uncertainty which is inappropriate for this procedure.

5 Discussion (conclusion(s) and evaluation)

The candidate was awarded **5 out of 8 marks**. The marks were awarded as follows:

5(a) **1 out of 1 mark** was awarded because the candidate has stated individual conclusions for each procedure.

5(b) **2 out of 3 marks** was awarded because the candidate has made a reasonable effort to discuss the accuracy and precision of their experimental

measurements. They did mention repetition although not range. They discussed limitation of equipment. The candidate mentioned systematic uncertainties for all procedures.

5(c) **1 out of 3 marks** was awarded because the candidate has discussed the interpretation of their results compared to the accepted value of g, although they misinterpreted their results for procedure two. The candidate discussed some problems that they had to overcome. They did not discuss the selection of their procedures. They did discuss further improvements but did not suggest any further work.

5(d) **1 out of 1 mark** was awarded because this is a good, competent project, well-worked through.

6 Presentation

The candidate was awarded **1 out of 2 marks**. The marks were awarded as follows:

6(a) **1 out of 1 mark** was awarded because the candidate has included an informative title page, contents page and page numbers.

6(b) **0 out of 1 mark** was awarded because although the candidate has cited at least three references, the reference style used is neither Harvard or Vancouver.

Overall

The candidate was awarded a total of **21 out of 30 marks**.

Candidate 3

1 Abstract

The candidate was awarded **0 out of 1 mark** because the candidate has stated aims but has not identified all methods or reported all findings (drop weight is missing). In addition, the finding of direct proportion for Jaeger's method is not supported by the candidate's graph.

2 Underlying physics

The candidate was awarded **2 out of 4 marks** because there is a reasonable attempt at a referenced derivation of the relationship used in the drop weight method. However, the candidate does not define all the symbols used and Figure 2.1 does not assist.

A more complete diagram would have been helpful for procedure 2, as the explanation refers to apparatus not shown in Figure 3.1. Most variables are defined but there are inconsistencies between the variables in the text and those in the accompanying diagram, and the use of (2t+2t) rather than (2l+2t).
There is a reference to, but no attempt to demonstrate understanding of, the underlying physics for procedure three. The candidate's use of non-standard symbols is confusing.

3 Procedures

The candidate was awarded **4 out of 7 marks**. The marks were awarded as follows:

3(a) **1 out of 2 marks** was awarded because the candidate has included a diagram for each procedure. Most labels are present, but some apparatus is missing from each of the diagrams.

3(b) **1 out of 2 marks** was awarded because there are a number of instances where the candidate has used an incorrect tense, therefore the maximum that can be awarded is 1 mark.

Despite some omissions of range/repetition, the descriptions are sufficient for replication.

3(c) **2 out of 3 marks** was awarded because the candidate has selected a range of procedures commensurate with Advanced Higher level.

The execution of procedure one lacks sophistication and could have been adapted to allow for graphical analysis by utilising a range of internal diameters, rather than a single glass tube.

Procedure three has been adapted from a direct measurement of γ to investigate the effect of temperature on surface tension, however no attempt was made to determine density of water by experiment.

4 Results (including uncertainties)

The candidate was awarded **1 out of 8 marks**. The marks were awarded as follows:

4(a) **0 out of 1 mark** was awarded because the candidate has not included raw data for procedure three.

4(b) **1 out of 4 marks** was awarded because the analysis for procedure one is based on the substitution of mean values into a relationship. There was an opportunity to perform graphical analysis by using tubing of different internal radii. The final value obtained for procedure one is correct, despite a missing $\times 10^{-3}$ in the penultimate line of the calculation.

Procedure two also relies on substitution of averaged values into a relationship. There is inappropriate averaging of the slide dimensions, which affects the individual values of surface tension obtained by the candidate.

In procedure three, the sample calculation of γ is incorrect, but the correct value appears in the table on page 19. Surface tension values (column four, rows two and three) of the table are incorrect. The graph is small, lacks minor gridlines and the data points are too large.

4(c) **0 out of 3 marks** was awarded because the candidate has shown an awareness of scale reading and random uncertainty but not mentioned calibration uncertainty.

In procedure one, the candidate has attempted combinations but has made too many mistakes, for example the candidate has correctly handled the uncertainty in the mass of one droplet but goes on to apply the same factor of ten to the uncertainty in radius.

The approach adopted in procedure two is inappropriate as measurements of different slides were used to calculate random uncertainty.

For procedure three, there is no raw data, so uncertainties cannot be checked.

5 Discussion (conclusion(s) and evaluation)

The candidate was awarded **2 out of 8 marks**. The marks were awarded as follows:

5(a) **0 out of 1 mark** was awarded because the candidate has only provided a conclusion for procedure one.

5(b) **1 out of 3 marks** was awarded because the candidate has included a suggestion to improve the reliability of procedure one and repeated comments on cleanliness of equipment, despite the procedures including cleaning of glassware.

The candidate identifies issues with 'zeroing' of the torsion balance in procedure two.

The candidate has attributed the result of procedure three to problems with an airtight seal and bubbles in the manometer. They have identified that a metre stick was not an appropriate tool for measurement in this procedure.

5(c) **1 out of 3 marks** was awarded because the candidate raised problems with vibrations due to people walking by.

The candidate compares the results of procedures one and two to the accepted value to determine the 'best' method. The candidate identified procedure two to be the most accurate and the easiest to set up.

The candidate suggested the use of a travelling microscope to improve the procedure.

5(d) **0 out of 1 mark** was awarded because this is not a quality project.

6 Presentation

The candidate was awarded **2 out of 2 marks**. The marks were awarded as follows:

6(a) **1 out of 1 mark** was awarded because the project has a structure that is easy to follow.

6(b) **1 out of 1 mark** was awarded because the candidate has included references in Vancouver style and has cited them in the text. There are sufficient citations in the text to be awarded the mark despite the lack of edition numbers for the books in the list.

Overall

The candidate was awarded a total of **11 out of 30 marks**.

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